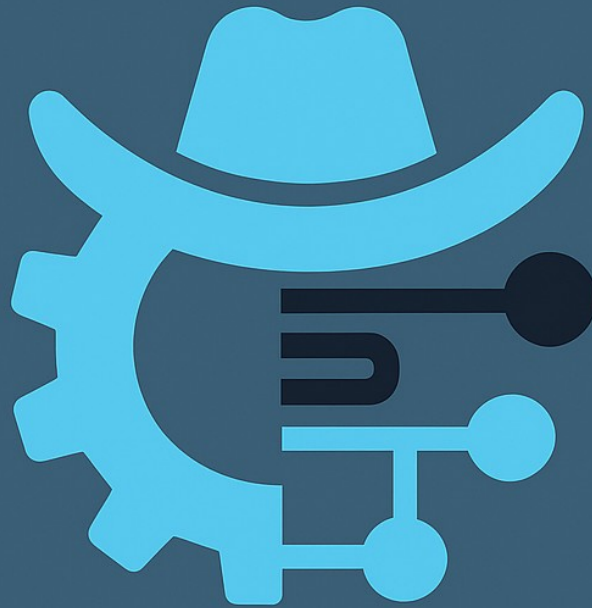




# OPEN SOURCE COWBOY CONSULTING

PIONEERING OPEN SOURCE FRONTIERS



## **The Open Replenishment Framework (ORF): Closing the Loop on Open Source Sustainability in Web3**

---

**Authored by:**

[Christian Taylor](#)

CoFounder & Chief Open Source Officer, Open Source Cowboy Consulting

**Contributors:**

[Terence “Tex” McCutcheon](#)

Senior Open Source Program Manager, Intersect

[Diane Mueller](#)

Managing Director, Research and Advisory Services, Bitergia

[Georg Link, Ph.D.](#)

Open Source Strategist, Bitergia; Co-Founder / Board member, CHAOSS Community

[Hart Montgomery](#)

CTO, Linux Foundation Decentralized Trust

[Heather Meeker](#)

Open Source Licensing Specialist & Co-Founder of the Chinstrap Community

[Andrew Aitken](#)

Web3 & AI Open Source Strategist; Founder of the Builder Bureau

**Date:**

August 18, 2026

**Organization:**

Open Source Cowboy Consulting

Web3 Open Source and Governance Advisory Firm

**Author's Note:** This paper is the third and final installment in a series and is meant to be read alongside the d(OSPO) paper and the OMF paper. The dOSPO defines the governance coordination layer (who decides); the OMF defines the maintenance deployment layer (how resources go out); the ORF defines the replenishment layer (how resources come back). Together they describe a complete closed loop, but each framework can also be used independently. The instruments catalogued herein describe existing models and options observable in production today, not “must use” mandates for the ORF to operate.



|                                                                              |           |
|------------------------------------------------------------------------------|-----------|
| <b>Glossary of Key Terms.....</b>                                            | <b>4</b>  |
| <b>Executive Summary.....</b>                                                | <b>6</b>  |
| <b>Framework Summary: The Complete Series.....</b>                           | <b>8</b>  |
| <b>1. The Problem: One-Way Funding and the Depletion Cycle.....</b>          | <b>10</b> |
| The Web3-Specific Dimension.....                                             | 11        |
| The Documented Costs of Depletion.....                                       | 11        |
| <b>2. The Replenishment Landscape: Existing Models and Their Limits.....</b> | <b>13</b> |
| Optimism — The Structural Revenue Model.....                                 | 13        |
| ENS — The Closest Closed-Loop Approximation.....                             | 14        |
| Tidelift — The Enterprise Demand Proof.....                                  | 14        |
| Protocol Guild — The Voluntary Pledge Commons.....                           | 14        |
| The Routing Mirage.....                                                      | 14        |
| <b>3. What the Landscape Is Missing.....</b>                                 | <b>15</b> |
| <b>4. Introducing the Open Replenishment Framework.....</b>                  | <b>17</b> |
| What ORF Is — and Is Not.....                                                | 17        |
| The ORF Thesis and the Five Economic Questions.....                          | 17        |
| Minimum Viable ORF.....                                                      | 18        |
| <b>5. Core Economic Principles.....</b>                                      | <b>19</b> |
| <b>6. Closed-Loop Architecture: How ORF Completes the Series.....</b>        | <b>20</b> |
| <b>7. The Two-Dimensional Instrument Taxonomy.....</b>                       | <b>22</b> |
| <b>8. The Five Revenue Families.....</b>                                     | <b>23</b> |
| <b>9. Governance and Institutional Separation.....</b>                       | <b>26</b> |
| <b>10. The Service Capacity Test.....</b>                                    | <b>28</b> |
| <b>11. Measuring Replenishment: The Six Formal Ratios.....</b>               | <b>29</b> |
| <b>12. The Eight Hard Gates for Self-Sustainability.....</b>                 | <b>31</b> |
| <b>13. Revenue Correlation Classes and True Diversification.....</b>         | <b>33</b> |
| <b>14. Evidence: The P0-P5 and D0-D5 Scales.....</b>                         | <b>35</b> |
| <b>15. Feasibility: The Tier 1 Scenario Model.....</b>                       | <b>37</b> |
| <b>16. Implementation Roadmap.....</b>                                       | <b>40</b> |
| <b>17. The Reporting Standard.....</b>                                       | <b>41</b> |
| <b>18. Failure Modes.....</b>                                                | <b>42</b> |
| <b>19. Research Status and the Road Ahead.....</b>                           | <b>43</b> |
| <b>20. Adversarial Review: Critiques and Rebuttals.....</b>                  | <b>44</b> |
| <b>Conclusion.....</b>                                                       | <b>46</b> |
| <b>About Open Source Cowboy.....</b>                                         | <b>47</b> |
| <b>Evidence Summary.....</b>                                                 | <b>47</b> |
| <b>Bibliography.....</b>                                                     | <b>49</b> |



## Glossary of Key Terms

| Term                                      | Definition                                                                                                                                                                                                                                             |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ORF</b>                                | Open Replenishment Framework. A portfolio design framework for identifying, validating, collecting, diversifying, measuring, and routing recurring sources of economic value back into the maintenance of shared open source infrastructure.           |
| <b>Replenishment</b>                      | The conversion of economic value enabled by open infrastructure into recurring, auditable net inflows that finance the ongoing cost of maintaining it. Distinct from funding (one-way disbursement) and from routing (moving already-collected money). |
| <b>Revenue Family</b>                     | One of five categories organizing all valid replenishment instruments: Structural Network Revenue, Enterprise Earned Revenue, Membership & Certification, Voluntary/Incentivized Contributions, and Capital Income.                                    |
| <b>Net Contribution</b>                   | Gross receipts minus all cost-to-collect: sales, service delivery, legal, tax, and administrative overhead. ORF evaluates every mechanism on net contribution, never gross revenue.                                                                    |
| <b>Cost Floor (C_base)</b>                | The empirically verified annual cost of maintaining an ecosystem's defined infrastructure scope. The denominator of every ORF coverage ratio.                                                                                                          |
| <b>Portfolio Coverage Ratio (PCR)</b>     | Total recurring non-inflationary net inflows divided by the cost floor. $PCR \geq 1.0$ means replenishment fully covers baseline maintenance.                                                                                                          |
| <b>Revenue Correlation Class</b>          | A grouping of inflows by shared economic risk exposure. True diversification requires multiple uncorrelated classes, not multiple instruments.                                                                                                         |
| <b>Counter-Value</b>                      | The independent, tangible benefit a payer receives in exchange for optional commercial payment — assurance, SLAs, certification, training. The legitimacy test for all non-structural collection.                                                      |
| <b>Structural Collection</b>              | Protocol-native revenue mechanisms (fee splits, sequencer contributions) where participation is not optional. Requires explicit network-governance legitimacy rather than counter-value.                                                               |
| <b>Fork-Resilient Economic Anchor</b>     | A non-copyable economic asset — maintainer capacity, canonical state, certification trust, enterprise relationships — that retains commercial value even when the underlying code is forked.                                                           |
| <b>Service Capacity Test</b>              | A formal five-condition test an ecosystem must pass before selling contractual maintenance guarantees (LTS/SLAs). Failing ecosystems are restricted to assurance products without remediation promises.                                                |
| <b>External Precedent Rating (P0-P5)</b>  | A scale describing how mature a mechanism is where it already operates: Conceptual, Emerging, Operating, Sustained, Scaled, Cycle-Tested. Answers “does this work somewhere?”                                                                          |
| <b>Deployment Evidence Rating (D0-D5)</b> | A separate scale describing local deployment proof within the adopting ecosystem: Hypothesis, Buyer Validated, Paid Pilot, Renewable, Scaled, Resilient. Answers “does this work here?”                                                                |
| <b>Mechanism Concentration</b>            | The largest single revenue mechanism's net contribution divided by total recurring net                                                                                                                                                                 |



| Term                        | Definition                                                                                                                                                                                |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ratio (MCR)</b>          | inflows. Distinct from RCR, which measures counterparty concentration. Reported and reviewed rather than hard-gated.                                                                      |
| <b>Routing Rail</b>         | Technical infrastructure (Drips, Superfluid) that moves already-collected money down the supply chain. Never counts as independent revenue.                                               |
| <b>Allocation Engine</b>    | An algorithm or voting system that determines recipient split percentages. Never counts as independent revenue.                                                                           |
| <b>Neutral Legal Entity</b> | The incorporated wrapper (Polkadot PCF pattern) that signs commercial contracts, invoices customers, files taxes, and carries liability — separated from governance and policy authority. |
| <b>Treasury Relabeling</b>  | The failure mode of calling existing grant or treasury transfers “revenue.” Blocked by ORF’s cash-evidence requirements.                                                                  |
| <b>Endowment Fantasy</b>    | The failure mode of treating capital yield as a bootstrap mechanism. Generating ~\$3M/year of spendable yield requires roughly \$60M–\$100M+ of productive principal.                     |
| <b>dOSPO</b>                | Decentralized Open Source Program Office. The governance coordination layer defined in the first paper of this series: who decides.                                                       |
| <b>OMF</b>                  | Open Maintenance Framework. The maintenance deployment layer defined in the second paper: how resources go out. ORF completes the loop: how resources come back.                          |



## Executive Summary

Open source software does not suffer from a lack of economic value. It suffers from a failure to connect the value created by open infrastructure with the resources required to maintain that infrastructure.

The first two papers in this series addressed governance and deployment. The Decentralized Open Source Program Office (dOSPO) defined a community-mandated coordination layer that separates policy authority from operational execution. The Open Maintenance Framework (OMF) defined the operational architecture that translates governance mandates and treasury capital into sustained maintainer programs. Both frameworks share an unstated dependency: capital must exist to deploy.

In nearly every Web3 ecosystem today, that capital arrives through one-way mechanisms — token issuance, treasury reserves, grants, donations, and corporate sponsorship — that deplete without replenishing. The infrastructure funded by that expenditure subsequently enables enormous commercial and network activity, yet little or none of the resulting value returns to the maintenance system that produced it. The result is a structural depletion cycle: every maintenance cycle begins with another search for finite capital.

This paper introduces the Open Replenishment Framework (ORF): an economic and governance framework designed to close that loop. ORF is not a single revenue mechanism. It treats sustainability as a portfolio design problem, organizing replenishment across five revenue families while strictly separating revenue generation from the routing rails and allocation engines that merely move money. The framework enforces measurement discipline through six formal replenishment ratios, eight hard gates for declaring self-sustainability, correlation-aware diversification requirements, a mandatory Service Capacity Test before contractual liability may be sold, and paired P0–P5 precedent and D0–D5 deployment scales that separate external precedent from local proof.

Critically, ORF does not propose monetizing open source code. The code remains open. Value is captured around the economic assets that cannot be reproduced by copying a repository: maintainer capacity, verified compatibility, canonical network activity, trusted certification, enterprise response commitments, and institutional coordination.

The evidence base is substantial but incomplete. Optimism demonstrates structural sequencer revenue at production scale. ENS demonstrates canonical service revenue feeding a governed endowment. Tidelift demonstrates enterprise willingness to pay for open infrastructure assurance. Protocol Guild demonstrates large-scale voluntary pledges. Polkadot demonstrates neutral legal execution beneath decentralized governance. Yet no reviewed ecosystem combines these mechanisms into a complete, measured, closed-loop architecture. ORF's contribution is that synthesis — presented as a Stage 0 research candidate, a testable architecture rather than a solved problem.

## What This Paper Contributes

|                                        |                                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Portfolio economic architecture</b> | A concrete framework for ecosystem replenishment — governance-agnostic, chain-agnostic, and organizationally neutral — that treats sustainability as portfolio design rather than a search for one revenue mechanism. |
| <b>Revenue / routing separation</b>    | ORF's core analytical contribution: strict decoupling of revenue sources from routing rails and allocation engines, ending the persistent confusion that counts payment infrastructure as sustainability.             |
| <b>Measurement discipline</b>          | Six replenishment ratios, six correlation classes, a twelve-field reporting standard, and eight hard non-compensable gates that together define what “self-sustaining” actually means.                                |
| <b>Liability safeguards</b>            | A formal Service Capacity Test gating the sale of maintenance guarantees, plus                                                                                                                                        |



|                  |                                                                                                                                                                            |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | Neutral Legal Entity separation that prevents commercial execution from capturing technical governance.                                                                    |
| Evidence honesty | Paired P0–P5 and D0–D5 scales separating “this works somewhere” from “this works here,” and a Stage 0 research designation that declines to overstate what ORF has proven. |

### Who This Document Is For

This framework document is written for Web3 ecosystem designers, DAO governance participants, treasury stewards, blockchain foundation operators, enterprise open source buyers, and community leaders responsible for the long-term financing of decentralized infrastructure. It is grounded in comparative analysis of real-world production systems and is designed to be adopted, tested, and falsified — not merely read.



## Framework Summary: The Complete Series

The dOSPO, OMF, and ORF together constitute a complete institutional model for governing, sustaining, and replenishing open source infrastructure in decentralized ecosystems. This page provides a 30-second orientation to the full system.

|                                       |                                                                                                                                                                                                   |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Governance Primitive<br/>dOSPO</b> | Who decides. A community-mandated coordination layer separating policy authority from operational execution: policy domains, governance bodies, operator mandates, and replaceability safeguards. |
| <b>Operational Framework<br/>OMF</b>  | How resources go out. The programmatic deployment engine: Maintainer Retainers, Code Bounties, Contributor Pathways, Operational Support, Incubation, and Resilience programs.                    |
| <b>Economic Framework<br/>ORF</b>     | How resources come back. The replenishment portfolio: five revenue families, net-contribution accounting, diversification rules, hard sustainability gates, and deployment evidence standards.    |
| <b>Revenue Families</b>               | Structural Network Revenue · Enterprise Earned Revenue · Membership & Certification · Voluntary & Pledges · Capital Income                                                                        |
| <b>Measurement Discipline</b>         | 6 Replenishment Ratios · 8 Hard Gates · 6 Correlation Classes · P0–P5 & D0–D5 Evidence Scales · 12-Field Reporting Standard                                                                       |
| <b>Legitimacy Safeguards</b>          | Counter-Value Requirement · Governance Authorization for Structural Collection · Service Capacity Test · Neutral Legal Entity Separation · Independent Audit                                      |



## The Architecture in One Diagram

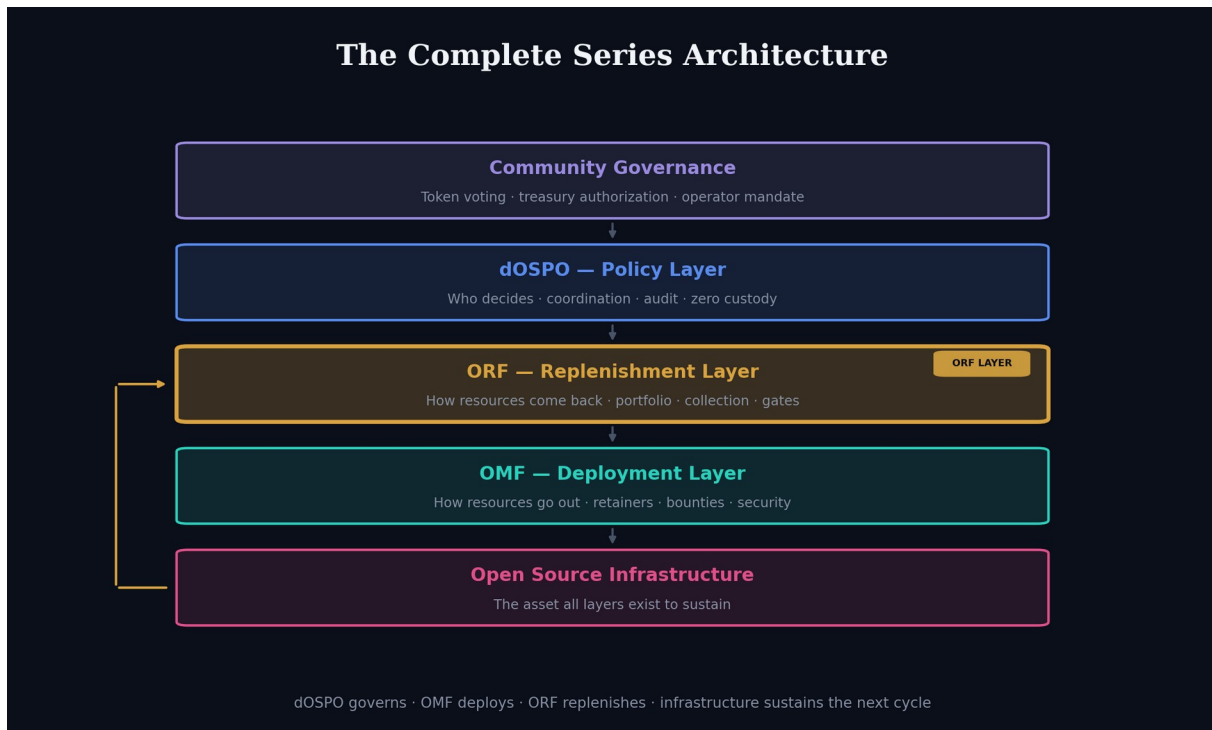


Figure 1: The Complete Series Architecture — dOSPO governs, OMF deploys, ORF replenishes.



## SECTION 1

## The Problem: One-Way Funding and the Depletion Cycle

Web3 ecosystems have built the most heavily capitalized funding mechanisms in the history of open source. Multi-billion-dollar treasuries, programmable disbursement, and transparent on-chain accounting represent genuine advances over the donation jars and corporate sponsorships of the previous era. Yet nearly all of this capital flows in one direction.

A typical ecosystem behaves approximately as follows. A treasury, foundation, or grant program disburses capital to maintainers and projects. Those maintainers sustain open infrastructure — protocol clients, SDKs, cryptographic libraries, developer tooling. That infrastructure enables economic and network activity: transactions, applications, enterprise adoption, exchange volume. The activity generates enormous external value.

And the flow stops there.

When the next maintenance cycle begins, the ecosystem must again locate capital from a treasury allocation, a token issuance event, a donor, or a new funding proposal. The value created by the infrastructure never returns to the system that maintains it. This model can fund open source. It does not sustain open source.

### RESEARCH FINDING

*Open source components appeared in 96% of the commercial codebases analyzed in Synopsys' 2024 Open Source Security and Risk Analysis. Separately, Hoffman, Nagle & Zhou estimate the supply-side replacement value of widely used open source software at approximately \$4.15 billion and its demand-side value at approximately \$8.8 trillion. The maintenance systems beneath that value remain structurally dependent on finite, one-way capital.*

— Synopsys OSSRA (2024); Hoffman, Nagle & Zhou, Harvard Business School (2024)

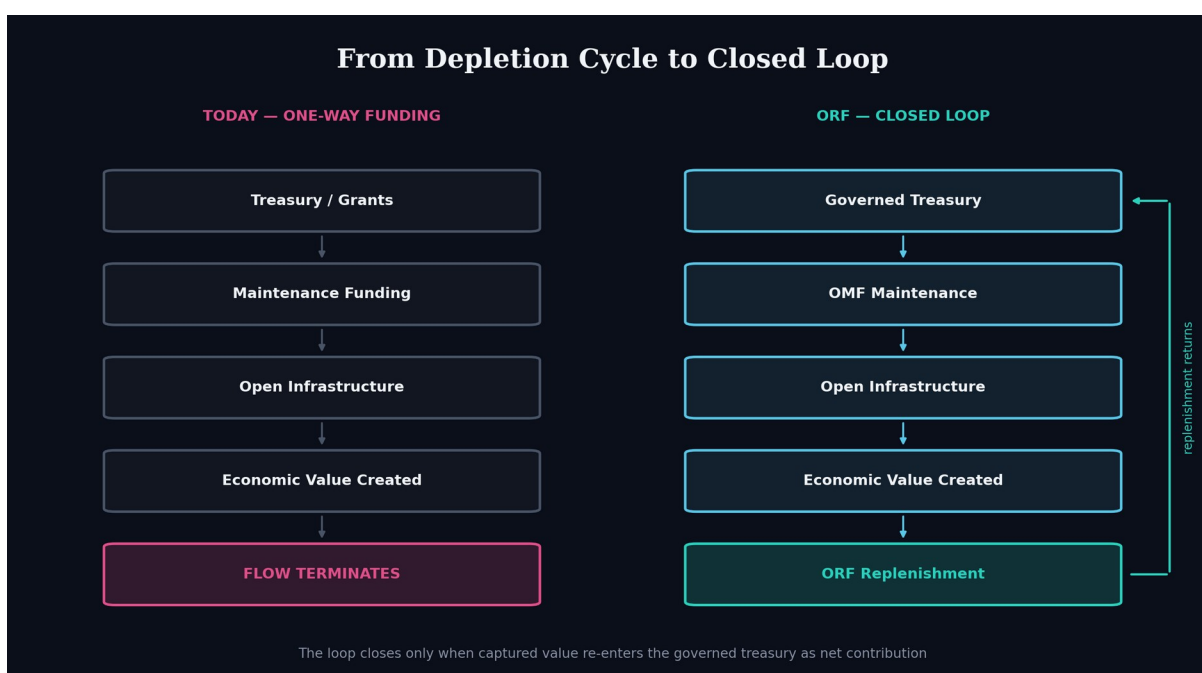




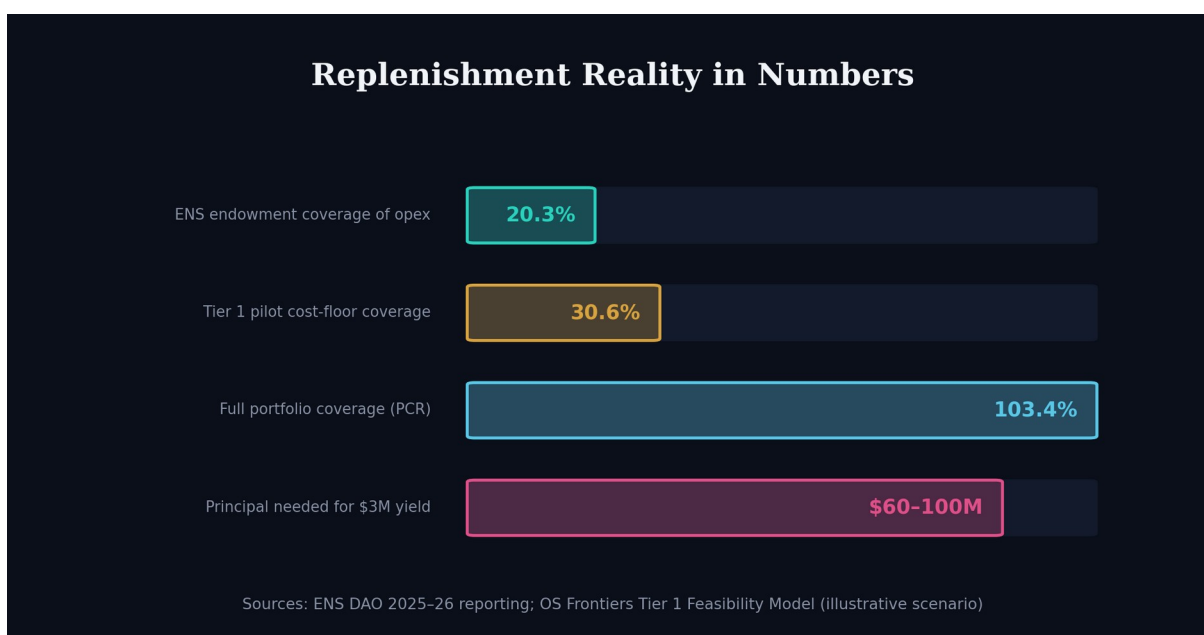
Figure 2: From Depletion Cycle to Closed Loop — the structural difference ORF addresses.

## The Web3-Specific Dimension

Web3 sharpens this problem in three ways. First, treasuries are denominated in native tokens, so the real-dollar value of committed maintenance funding is correlated with the very market cycles that determine ecosystem stress — funding capacity collapses precisely when resilience is most needed. Second, monetary expansion disguises the problem: newly issued tokens allocated toward maintenance feel like revenue but represent dilution of existing holders, not captured external value. Third, the visible success of a handful of protocol-fee models has created a false confidence that structural revenue alone solves sustainability — ignoring correlation risk, payer concentration, and the governance legitimacy such mechanisms require.

## The Documented Costs of Depletion

|                                         |                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Treasury Drawdown Without Return</b> | Ecosystem treasuries disburse hundreds of millions annually into maintenance and grants with no mechanism for value recapture. Every allocation is a permanent depletion event against a finite reserve.                                                                                                                            |
| <b>Token-Related Collapse</b>           | Maintenance budgets denominated in native tokens can lose half their purchasing power between governance approval and disbursement. A 50%+ drawdown is a routine crypto market event, not a tail risk.                                                                                                                              |
| <b>Payer Concentration Shock</b>        | Optimism's Superchain revenue model — the strongest structural precedent in Web3 — was materially exposed when Base moved toward a Base-managed unified software stack in 2026 while maintaining a transitional support relationship, demonstrating that even protocol-level revenue carries counterparty exit and dependency risk. |
| <b>The Endowment Mirage</b>             | Capital yield is repeatedly proposed as the answer, yet generating ~\$3M/year of spendable income at 3–5% real yield requires \$60M–\$100M+ of productive principal. ENS — the best-capitalized endowment in the space at \$93.4M AUM — covers roughly one-fifth of annual operating expenses from returns.                         |





*Figure 3: Replenishment reality in numbers — verified coverage figures against modeled targets.*

The series diagnosis therefore extends one level deeper than the OMF paper's conclusion. The OMF established that the sustainability crisis is not primarily a funding-availability problem but an operational-architecture problem. The ORF adds the economic corollary: even a perfectly operated deployment architecture depletes without a replenishment architecture behind it. Governance without deployment is inert; deployment without replenishment is terminal.



## The Replenishment Landscape

A meaningful set of production mechanisms now exists that captures economic value for open source ecosystems from different angles — structural protocol revenue, canonical services, enterprise subscriptions, voluntary pledges, and capital management. Understanding these mechanisms, their strengths, and their gaps is essential context for what ORF adds. The following comparative analysis draws on primary governance records, financial disclosures, and protocol documentation from each initiative.

### Overview of Major Replenishment Precedents

| Initiative                          | Revenue Model                                                                | Verified Scale                                                                                                 | Key Risk                                                              | Loop?   |
|-------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------|
| <b>Optimism Superchain</b>          | Greater of 2.5% of Chain Revenue or 15% of Net Onchain Profit from OP Chains | Reported cumulative Superchain revenue and Retro Funding at scale; 50% revenue-linked OP buyback approved 2026 | Payer concentration; Base stack transition (exit and dependency risk) | Partial |
| <b>ENS DAO / Endowment</b>          | Canonical .eth registration fees + governed endowment (May 2026 EP6.46 IPS)  | \$18.22M 2025 revenue; ~\$93.4M AUM; >\$8M net DeFi returns since inception                                    | Single revenue source; yield covers ~20.3% of expenses                | Closest |
| <b>Tidelift (acquired by Sonar)</b> | Enterprise assurance subscriptions; SBOM-based maintainer payments           | Named customers: Cisco, Fannie Mae, U.S. Air Force                                                             | Requires sales, support, and legal capacity most DAOs lack            | Partial |
| <b>Protocol Guild</b>               | Voluntary 1% token/yield pledges into vesting split contract                 | \$7.2M from 6,202 donors (2025); 196 members as of Q2 2026; \$80M+ committed                                   | Voluntary contributions can stop being voluntary                      | Partial |
| <b>Polkadot / PCF</b>               | Protocol fees, slashes, issuance; Cayman executor; Anemoy yield              | ~\$70.6M 2025 treasury spend; \$1.5M Anemoy allocation                                                         | Issuance-heavy inflows; token correlation                             | Partial |
| <b>Octant (Golem)</b>               | Staking yield from locked capital funds public goods                         | 100,000 ETH staked; ~460 ETH (~\$1.7M) Epoch 8                                                                 | 100k ETH bootstrap barrier                                            | No      |
| <b>Nouns DAO</b>                    | Perpetual daily NFT auction; 100% proceeds to treasury                       | One Noun auctioned every 24h indefinitely                                                                      | Depends on scarce cultural asset demand                               | No      |
| <b>Cardano POSM</b>                 | Treasury-funded maintenance programs (20% of fees to treasury)               | 5.885M ADA 2025 budget; \$300K bounty pool fully utilized                                                      | Deployment precedent, not replenishment                               | No      |

### Optimism — The Structural Revenue Model

Optimism's Superchain represents the strongest native protocol-revenue precedent in Web3. In Optimism's own terms, Standard OP Chains contribute the greater of 2.5% of Chain Revenue or 15% of Net Onchain Profit to the Optimism Collective — a mechanism that has financed cumulative Superchain revenue and a large Retro Funding programme, and that governance extended in 2026 by approving the routing of 50% of revenue to OP



buybacks. The critical negative lesson arrived the same year: Base moved toward a Base-managed unified software stack while maintaining a transitional support relationship with Optimism, demonstrating that a single dominant chain contributor creates exit and dependency risk even inside a “structural” mechanism. Structural does not mean diversified. The exact cumulative ETH, OP, and grantee totals widely repeated in ecosystem commentary should be sourced to a dated Optimism publication before republication; this paper relies on the mechanism and the governance record rather than those aggregates.

## ENS — The Closest Closed-Loop Approximation

ENS is the closest thing the industry has to a working closed loop. Canonical .eth registration revenue (\$18.22M in 2025) flows automatically from the registrar into DAO control. Under the May 2026 EP6.46 Investment Policy Statement, the non-custodial endowment — executed by a professional manager (KPK) within governed boundaries — reported approximately \$93.4M in assets under management and more than \$8M in net DeFi returns since inception, and those returns finance operations. The instructive limitation: endowment yield covered approximately 20.3% of 2025 operating expenses. Even the best-capitalized capital sleeve in Web3 open source functions as a resilience layer, not a foundation.

## Tidelift — The Enterprise Demand Proof

Tidelift, acquired by Sonar, provides the strongest evidence that enterprises will pay recurring subscription revenue for assurance around open infrastructure — dependency intelligence, vulnerability interpretation, verified maintenance status — and that this revenue can flow upstream to maintainers via SBOM-based allocation. Its named customers include Cisco, Fannie Mae, and the U.S. Air Force. Its limitation as a template is operational: it required a venture-backed Delaware corporation with professional sales, support, and legal capacity — exactly the institutional apparatus most Web3 ecosystems lack, and that ORF's Neutral Legal Entity architecture is designed to supply.

## Protocol Guild — The Voluntary Pledge Commons

Protocol Guild demonstrates coordinated voluntary replenishment at meaningful scale: \$7.2M raised from 6,202 unique donors during 2025, 196 members as of its Q2 2026 quarterly audit, and more than \$80M committed by major ecosystem projects. The OMF paper analyzed the Guild as a continuity-funding deployment precedent; from the ORF side it is a Family D contribution source — valuable, capture-resistant, and structurally revocable. Donations that flow because of goodwill can stop flowing for the same reason. This is diversification material, not a foundation for guaranteed operating liabilities.

## The Routing Mirage — Drips, Superfluid, and Allocation Engines

Drips (dependency-tree splits), Superfluid (per-second streaming), and allocation experiments such as Deep Funding (\$220K 2025 challenge across 34 seed repos and 5,000+ dependencies) are frequently cited as sustainability solutions. They are excellent infrastructure — and they generate zero replenishment. They move and divide money that something else must first collect. The persistent confusion between routing and revenue is one of the most common analytical failures in ecosystem sustainability discussions, and ORF's taxonomy exists in large part to eliminate it.



## What the Landscape Is Missing

Taken together, the precedents surveyed in Section 2 represent real and meaningful progress. Structural fee routing, canonical service revenue, enterprise assurance, voluntary pledge commons, and governed endowments all work — somewhere, for someone. But each captures one slice of the replenishment problem, and none provides a comprehensive, measured, portable architecture that any ecosystem can adopt.

### The Gaps in the Current Landscape

|                                       |                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>No portfolio architecture</b>      | Among the precedents reviewed, each is a single-mechanism story. ENS has canonical fees plus capital. Optimism has sequencer revenue. Tidelift has enterprise subscriptions. This review did not identify an ecosystem operating a deliberately diversified portfolio across uncorrelated revenue classes with measured coverage targets. |
| <b>Gross-revenue thinking</b>         | Public discussion of ecosystem revenue almost universally cites gross figures. Sales cost, service delivery, legal, tax, insurance, and administrative overhead are rarely disclosed, making unprofitable programs indistinguishable from sustainable ones.                                                                               |
| <b>Revenue conflated with routing</b> | Streaming rails, dependency splits, and AI allocation engines are routinely presented as sustainability mechanisms. They distribute; they do not replenish. To our knowledge, no published framework enforces this separation as a first-class design constraint.                                                                         |
| <b>Issuance conflated with income</b> | Token issuance directed at maintenance is habitually counted as ecosystem revenue. It is dilution — a transitional funding source that cannot appear in any honest self-sustainability calculation.                                                                                                                                       |
| <b>No sustainability test</b>         | Ecosystems declare themselves sustainable on the basis of treasury size, current fee revenue, or a single large customer. This review did not identify a published framework defining hard, non-compensable conditions that must all hold simultaneously.                                                                                 |
| <b>Precedent mistaken for proof</b>   | The existence of a mechanism somewhere — CNCF certification, Tidelift subscriptions, Optimism fee splits — is treated as evidence it will work locally. Among the frameworks reviewed, none separates external precedent from local deployment evidence on distinct scales.                                                               |

*The common failure across existing approaches is economic: collection mechanisms exist, but portfolio architectures do not. Ecosystems have built individual revenue experiments, but not systems capable of measuring whether the resulting inflows actually finance the cost of maintenance under stress. The Open Replenishment Framework addresses this missing layer — defining the economic architecture through which ecosystem value becomes measured, diversified, governed replenishment.*



## The Synthesis ORF Provides

The landscape reveals a clear pattern: multiple ecosystems have independently converged on fragments of the same economic logic — fee routing toward shared treasuries, enterprise assurance products, membership consortia, pledge commons, governed capital. Optimism's sequencer split, ENS's registrar flow, Tidelift's subscriptions, Protocol Guild's vesting pledges, and the Linux Foundation's membership tiers all reflect partial answers to the same question: how does value come back?

ORF synthesizes these convergent fragments into a single portable architecture. It is not derived from any single precedent — it is the generalization of patterns that multiple independent production systems have validated, joined to the measurement and governance discipline that, among the systems reviewed, none of them individually enforces. Any Web3 ecosystem, regardless of chain, treasury mechanism, or governance structure, can implement ORF without dependency on or license from any specific prior program.



## Introducing the Open Replenishment Framework

The Open Replenishment Framework (ORF) is a governance and economic design framework for building an ecosystem replenishment portfolio. It functions as the economic architecture of the series — connecting the value created by open infrastructure back to the governed treasury that the dOSPO oversees and the OMF deploys.

ORF is designed to stand alone. It can be executed by foundations, DAOs, elected committees, or independent operators — without dependency on any specific governance structure, chain, or treasury mechanism. Where ecosystems have adopted the complementary series frameworks, ORF operates as the revenue counterpart to the OMF beneath dOSPO policy coordination. Where they have not, ORF deploys directly through whatever accountable governance authority the ecosystem already possesses.

### What ORF Is — and Is Not

|                   |                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ORF IS</b>     | A portfolio design framework for ecosystem replenishment. A taxonomy separating revenue from routing and allocation. A measurement discipline of ratios, gates, correlation classes, and evidence scales. A governance architecture separating policy from commercial execution. A framework any ecosystem can adopt without precondition.                                  |
| <b>ORF IS NOT</b> | A proposal to restrict open source licensing. A requirement to charge for code access. A claim that donations are inherently unsustainable, or that protocol fees alone can sustain every ecosystem. An investment product. A replacement for governance. A mechanism for commercial buyers to purchase technical control. Dependent on dOSPO, OMF, or any other framework. |

The framework instead directs ecosystems toward economic assets that cannot be reproduced by copying a repository: maintainer capacity, verified compatibility, canonical network activity, trusted certification, vulnerability intelligence, recognized trademarks, enterprise response guarantees, and established coordination. This distinction is fundamental: the code remains open; value is captured around services, assurances, economic activity, coordination, and institutional trust.

### The ORF Thesis and the Five Economic Questions

#### CORE THESIS

*An open source ecosystem becomes financially resilient when a sufficient portion of the economic value enabled by its infrastructure can be converted into diversified, recurring, auditable net inflows capable of financing the ongoing cost of maintaining that infrastructure.*

— ORF Framework Definition

This differs from simply maximizing treasury revenue. Every candidate replenishment mechanism must answer five questions before it enters the portfolio:

- Origin — Where does the economic value originate?



- ▶ Legitimacy — Why would a participant legitimately pay or contribute?
- ▶ Cost — What does collection actually cost?
- ▶ Risk — How exposed is the mechanism to market, customer, governance, or token risk?
- ▶ Coverage — Does enough net value return to maintenance to materially reduce dependence on finite treasury capital?

A mechanism that cannot answer all five is not a replenishment instrument; it is a hope.

## Minimum Viable ORF

For portability to be meaningful, the minimum viable implementation must be precisely defined. An ecosystem has implemented ORF if it has all five of the following:

|                                          |                                                                                                                                                                                                                                |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Verified cost floor</b>            | The ecosystem has empirically established its annual maintenance cost floor — the actual cost of maintaining its defined infrastructure scope — and published the methodology.                                                 |
| <b>2. Classified inflow inventory</b>    | All current inflows are classified under the ORF taxonomy: revenue source, contribution source, issuance, routing rail, allocation engine, or capital management. Issuance and routing are excluded from replenishment totals. |
| <b>3. At least one earned instrument</b> | At least one commercial or structural instrument is collecting real receipts — an assurance subscription, membership tier, certification program, or governance-legitimized fee mechanism — measured on net contribution.      |
| <b>4. Net-contribution reporting</b>     | Quarterly public reporting covers gross receipts, itemized cost-to-collect, net contribution to the governed treasury, payer concentration, and outstanding liabilities.                                                       |
| <b>5. Ratio and gate review cadence</b>  | The six replenishment ratios are computed at least annually and reviewed against the eight hard gates. Coverage gaps and correlation exposure drive the next portfolio decision, not legacy commitments.                       |

An ecosystem meeting these five conditions is running ORF — even with a Portfolio Coverage Ratio far below 1.0. The framework measures honest progress toward self-sustainability; it does not require self-sustainability to begin.



## Core Economic Principles

ORF is grounded in five economic principles. These function as design constraints — defining what counts as replenishment, what does not, and how competing pressures must be navigated. Every instrument in the portfolio should be evaluated against them.

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Legitimacy &amp; Counter-Value</b>      | Optional commercial collection must provide independent, tangible counter-value to the payer: assurance, SLAs, certification, registry placement, training. Structural protocol collection is different in kind — participation may not be optional — and therefore requires explicit governance legitimacy from token holders and network stakeholders. ORF never blurs the two: instruments are either sold (counter-value) or governed (structural mandate), never quietly taxed. |
| <b>Net Contribution Over Gross Revenue</b> | Every mechanism is evaluated on net contribution after all cost-to-collect: collection, delivery, legal, tax, and administration. A program generating \$1,000,000 gross against \$900,000 of sales, legal, engineering, insurance, and support expense contributes \$100,000 of replenishment. ORF evaluates the \$100,000 — not the \$1,000,000.                                                                                                                                   |
| <b>Strict Functional Separation</b>        | Revenue sources (new money generated) are strictly decoupled from routing rails (smart contracts moving existing money) and allocation engines (algorithms or votes deciding splits). A system that moves \$1M from one wallet to another has improved distribution; it has not generated \$1M of replenishment.                                                                                                                                                                     |
| <b>Correlation-Aware Diversification</b>   | Three mechanisms are not three revenue streams. Protocol fees, staking income, validator pledges, and token-denominated contributions can all collapse simultaneously in a crypto contraction. Diversification is measured across economic correlation classes, not counted across instruments.                                                                                                                                                                                      |
| <b>Fork-Resilient Economic Anchors</b>     | An open repository can be forked; the economic system around it usually cannot. Maintainer knowledge, enterprise relationships, certification trust, trademarks, canonical state and liquidity, operational response capacity, and governance legitimacy are anchors that retain value under forking. ORF monetizes those anchors rather than manufacturing scarcity around code.                                                                                                    |

### PRINCIPLE PRECEDENT

*Value capture that survives forking attaches to non-copyable assets: official registry recognition, SLA maintenance commitments backed by real maintainer time, and canonical state and liquidity networks. A fork copies software lines; it does not copy human commitments, network effects, or institutional trust.*

— OS Frontiers Fork Resistance Analysis; ENS registrar economics; Tidelift enterprise model



## Closed-Loop Architecture

Open ecosystems adopting the full series operate a six-stage loop. Understanding where ORF sits — and what it explicitly does not do — is essential to understanding why the separation of layers matters.

### The Six Stages of the Loop

| Stage                              | Layer                       | Function                                                                                                                                                |
|------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Value Generation</b>         | Economy                     | Protocol activity, enterprise demand, membership, contributions, and capital returns create gross economic value around the infrastructure.             |
| <b>2. ORF Collection</b>           | ORF Operator + Legal Entity | Instruments collect gross receipts; sales, delivery, legal, tax, and reserve costs are deducted; net contribution transfers to the governed treasury.   |
| <b>3. Governed Treasury</b>        | Governance + dOSPO          | Net proceeds accumulate under dOSPO budget caps, capital policy, and reserve rules. Governance — not the ORF operator — decides deployment.             |
| <b>4. Routing &amp; Allocation</b> | Rails & Engines             | Approved allocations move through transparent rails (Drips splits, Superfluid streams) and allocation mechanisms. These distribute; they never collect. |
| <b>5. OMF Deployment</b>           | OMF Operator                | Capital becomes maintenance: retainers, bounties, pathways, security programs, incubation — per the OMF program architecture.                           |
| <b>6. Sustained Infrastructure</b> | Maintainers                 | Maintained infrastructure enables the next cycle of economic activity — returning the loop to Stage 1.                                                  |



## The ORF Operating Cycle

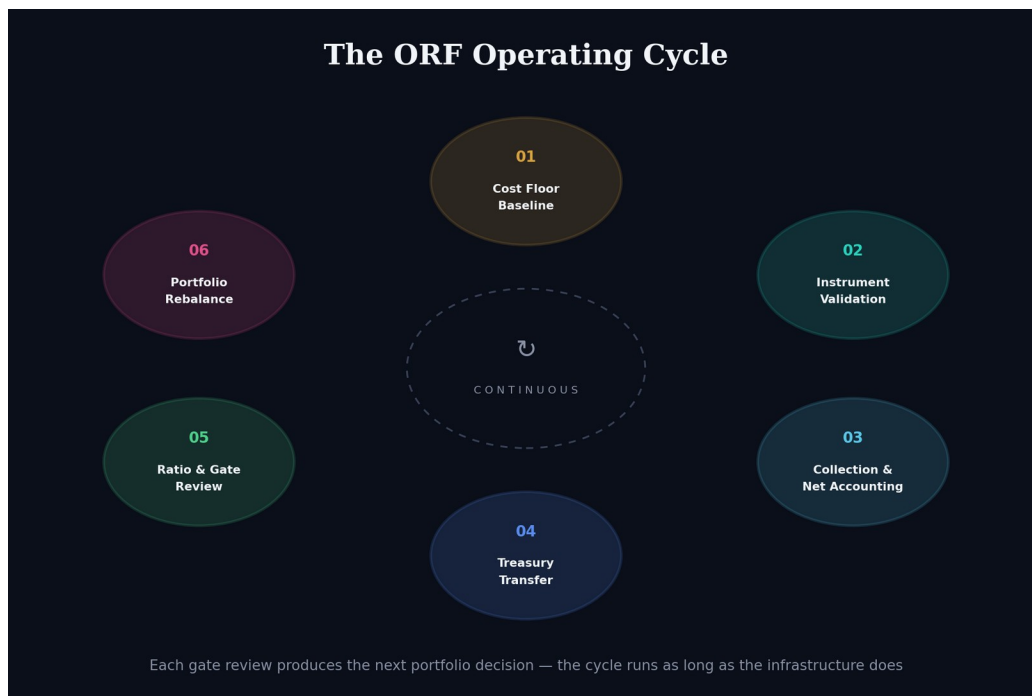


Figure 4: The ORF Operating Cycle — a recurring governance cadence, not a one-time exercise.

## The Most Important Architectural Rule

The load-bearing distinction in this architecture is the separation between generation, collection, allocation, and deployment. The dOSPO paper established that coordination must not become control. The OMF paper established that funding must not become ownership. The ORF corollary: distribution must not be mistaken for replenishment. Routing technologies and allocation engines — however sophisticated — occupy Stage 4 of the loop. Counting them as revenue is the accounting equivalent of counting a conveyor belt as a factory.

*ORF is not a revenue mechanism — it is a portfolio architecture. It defines how an ecosystem identifies legitimate sources of recurring value, measures their true net contribution, diversifies them across risk classes, and determines whether the resulting system actually finances the cost of maintenance. It performs this function with or without the other series frameworks sitting alongside it.*



## The Two-Dimensional Instrument Taxonomy

To eliminate conceptual confusion, every ORF instrument is explicitly classified across two dimensions. This prevents fundamentally different mechanisms from being grouped together simply because they involve money — the analytical failure at the root of the routing mirage.

### Dimension 1: Value-Origin Layer

|                          |                                                                         |
|--------------------------|-------------------------------------------------------------------------|
| <b>Protocol Layer</b>    | On-chain network activity, sequencer fees, canonical protocol services. |
| <b>Application Layer</b> | User-facing dApps, ecosystem tooling, dependency distribution.          |
| <b>Enterprise Layer</b>  | Commercial adopters, corporate users, institutional buyers.             |
| <b>Capital Layer</b>     | Accumulated treasury reserves, yield strategies, financial assets.      |
| <b>Delegation Layer</b>  | Validator networks, stake pools, community pledges.                     |

### Dimension 2: Functional Instrument Type

| Type                            | Function                                                       | Counts as Replenishment?          |
|---------------------------------|----------------------------------------------------------------|-----------------------------------|
| <b>Revenue Source</b>           | Generates or redirects new recurring money into sustainability | Yes                               |
| <b>Contribution Source</b>      | Voluntary or incentive-based capital transfers                 | Yes (revocability discount)       |
| <b>Capital Management</b>       | Generates financial return from accumulated principal          | Yes (yield only, never principal) |
| <b>Issuance Source</b>          | Monetary expansion allocated toward maintenance                | No — dilution, not captured value |
| <b>Routing Rail</b>             | Moves already-collected money down the supply chain            | No                                |
| <b>Allocation Mechanism</b>     | Determines recipient split percentages                         | No                                |
| <b>Financial / Risk Product</b> | Contingent claims, underwriting, repayment rights              | Research horizon (D0–D1)          |

Every instrument entry additionally carries two ratings on two deliberately separate scales: an External Precedent Rating (P0–P5), describing how mature the mechanism is where it already operates, and a Deployment Evidence Rating (D0–D5), describing what the adopting ecosystem has actually proven locally. The scales are distinct because they measure different things and have different criteria — external maturity can be established from public operation and disclosure, whereas local maturity requires the ecosystem's own receipts, renewals, and net contribution. CNCF-style conformance certification is P4 as an external precedent and D0 for any ecosystem that has never run a conformance program.



## The Five Revenue Families

All valid replenishment instruments are organized under five revenue families. Each family is validated by real-world precedents from the comparative landscape, and each carries distinct governance requirements, capacity requirements, and risk characteristics. Families can be implemented independently; mature portfolios combine several.

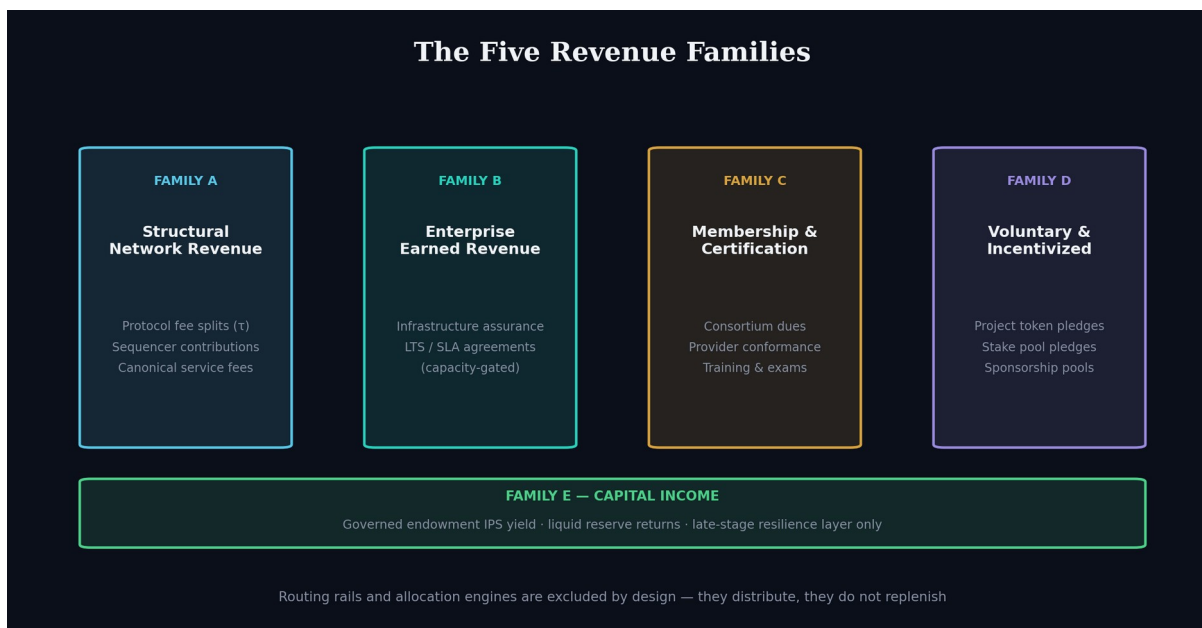


Figure 5: The Five Revenue Families — the complete replenishment portfolio taxonomy.

### Family A — Structural Network Revenue

Structural network revenue originates from native economic activity occurring inside the protocol itself: protocol transaction-fee routing (a fixed split of L1 fees to the treasury), sequencer profit contributions, and canonical protocol service fees.

| Instrument                           | External Precedent (P-scale)                                                       | Local Readiness                                           |
|--------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>Protocol Fee Routing</b>          | Cardano treasury cut; Polkadot fee split — P4 Scaled (L1)                          | Medium transferability; requires governance authorization |
| <b>Sequencer Profit Contribution</b> | Optimism: greater of 2.5% Chain Revenue or 15% Net Onchain Profit — P4 Scaled (L2) | High for rollup ecosystems                                |
| <b>Canonical Service Fees</b>        | ENS .eth registration, \$18.22M 2025 revenue — P4 Scaled                           | Low / contextual; requires a canonical namespace          |

Structural revenue has a decisive advantage: it scales automatically with network use, requiring no sales force and no per-customer relationship. It also carries the heaviest governance burden. A protocol fee split is not a customer purchase — participation may not be optional — so it changes how network economics are



distributed and therefore requires explicit governance legitimacy and transparent economic analysis before activation. This is precisely the class of decision the dOSPO's policy domains exist to route.

The Cardano treasury cut is worth stating precisely, because it illustrates the taxonomy catching a real-world mixed inflow. Cardano directs 20% of the epoch reward pot to the treasury, but that pot combines transaction fees with monetary expansion. Under ORF accounting, only the fee-derived portion qualifies as non-inflationary structural revenue; the monetary-expansion component is classified as issuance and excluded from the Portfolio Coverage Ratio. An ecosystem citing “a 20% treasury cut” as replenishment must therefore decompose the pot before claiming coverage.

#### CRITICAL POLICY NOTICE — ISSUANCE IS NOT REVENUE

*Monetary expansion allocated toward maintenance represents token dilution, NOT new economic revenue. It may legitimately finance maintenance as a transitional source. It CANNOT count toward non-inflationary self-sustainability metrics under any ORF calculation. Where a single mechanism blends fees and issuance — as the Cardano reward pot does — the inflow must be decomposed and only the fee-derived share counted.*

— ORF Instrument Catalog, Instrument A.4; Cardano monetary policy documentation

### Family B — Enterprise Earned Revenue

Enterprise earned revenue converts commercial demand around open infrastructure into recurring contract revenue. ORF defines two principal products separated by liability profile.

|                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Product A</b><br><b>Open Infrastructure Assurance</b>    | Commercial subscribers pay annual fees (indicatively \$25K–\$100K+) for risk reduction without buying maintainer control: maintained-project status, vulnerability interpretation, direct escalation channels, planned-change briefings, quarterly risk reports, and verified sustainer listing. Service capacity requirement is low-to-medium; no 24/7 patching is promised. The buyer purchases risk intelligence, not governance. Precedent: Tidelift — P4 externally, but typically D1 Buyer Validated locally; high transferability. |
| <b>Product B</b><br><b>Extended Lifecycle Support (LTS)</b> | High-value contracts (\$75K–\$250K+) promising defined backport windows (24/36 months), severity-graded acknowledgement times, and dedicated resolution paths. Precedents: Red Hat ELS, Canonical Ubuntu Pro. This product creates real contractual liability and is prohibited until the ecosystem passes the formal Service Capacity Test. For generic DAOs without contracted support organizations, this instrument is honestly rated D0 Hypothesis.                                                                                  |

***Do not sell maintenance guarantees that the ecosystem lacks the operational capacity to fulfill. Selling response commitments before establishing maintainer capacity converts a sustainability program into an unfunded liability.***

The OMF connection is direct: Product B is only sellable because OMF Maintainer Retainer programs exist to contract the capacity behind it. ORF sells what OMF sustains.

### Family C — Ecosystem Membership, Certification, and Training

Family C captures value from the ecosystem's institutional and knowledge layer without reducing access to the software.



| Instrument                                       | Model                                                                                                                                                                                                                               | External Precedent                                   |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| <b>Sustaining Consortium Membership</b>          | Tiered annual dues (\$10K–\$250K+): Supporter, Sustainer, Strategic Sustainer. Members receive working-group participation, briefings, events, and recognition. Technical governance remains 100% independent of membership status. | Linux Foundation membership — P4 Scaled              |
| <b>Certified Ecosystem Provider</b>              | Providers pay annual fees (\$10K–\$50K) to submit to objective conformance test suites, earning trademark usage, registry listing, and referral routing.                                                                            | CNCF Certified Kubernetes — 90+ offerings; P4 Scaled |
| <b>Professional Certification &amp; Training</b> | Proctored exams (\$300–\$750) and training bundles where employer demand exists.                                                                                                                                                    | Linux Foundation CKA/CKAD — P3 Sustained             |

**KEY PRINCIPLE**

*Payment buys participation and testing services. Certification requires passing objective technical standards. Payment MUST NOT purchase a passing result — a certification program that sells outcomes destroys the trust asset it monetizes.*

**Family D — Voluntary and Incentivized Contributions**

Family D includes Protocol Guild-style 1% token/yield pledges, protocol revenue pledges, validator stake-pool contributions, ecosystem sponsorships, and community public-goods pools. Protocol Guild's verified scale — \$7.2M from 6,202 donors in 2025, \$80M+ in project commitments — demonstrates that coordinated voluntary replenishment can finance significant maintainer support.

The structural weakness is equally demonstrated: voluntary contributions can stop being voluntary contributions. ORF therefore treats Family D as valuable diversification and community-alignment material, but a weak foundation for guaranteed operating liabilities. A maintenance organization should not sign multi-year obligations against revenue that contributors can discontinue without equivalent contractual commitment. Cardano Mission-Driven Pools and POSM community pools provide the Delegation-Layer analog (P2 Operating externally; D2 Paid Pilot locally; medium transferability).

**Family E — Capital Income**

Well-capitalized ecosystems may place reserves under a formal Investment Policy Statement targeting 3–5% real yield from low-risk assets (US Treasuries, stablecoins, ETH staking). External precedents: ENS Endowment (EP6.46; \$93.39M AUM; >\$8M cumulative net yield; professional execution by KPK within governed boundaries) and Octant (100,000 ETH staked; ~460 ETH distributed in Epoch 8).

**CRITICAL REALITY CHECK**

*At 3–5% real yield, generating ~\$3M/year of spendable income requires roughly \$60M–\$100M+ of productive principal — before operating costs, reserves, taxes, volatility, or inflation. Capital income is a late-stage resilience layer, NOT a Day-1 bootstrap solution. ENS — the best precedent available — covers approximately 20.3% of operating expenses from endowment returns.*

— ORF Instrument Catalog E.1; ENS DAO financial reporting, 2025–2026



## Governance and Institutional Separation

A sustainable replenishment model can still fail if economic authority concentrates in the wrong institution. The dOSPO paper established the general principle — coordination must not become control — and this section applies it to money.

### The Separated Roles

| Body                        | Authority                                                                                                                                | Explicitly Cannot                                                                             |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Community Governance</b> | Sets legitimate ecosystem policy; authorizes structural collection; approves budget caps; appoints and replaces operators                | Execute contracts; manage customers                                                           |
| <b>dOSPO</b>                | Coordinates policy, standards, measurement, and oversight across OMF and ORF operators; audits performance                               | Hold treasury custody; act as sales agent; assume commercial liability                        |
| <b>ORF Operator</b>         | Executes the replenishment strategy: product design, pipeline, pricing, collection oversight                                             | Set technical direction; allocate funds outside approved programs; self-authorize instruments |
| <b>OMF Operator</b>         | Executes maintenance deployment per the OMF program architecture                                                                         | Collect revenue; sign customer contracts                                                      |
| <b>Neutral Legal Entity</b> | Signs commercial SLAs and vendor contracts; invoices and collects; files taxes; maintains insurance; routes net proceeds to the treasury | Set policy; retain surplus beyond authorized reserves                                         |
| <b>Governed Treasury</b>    | Receives net proceeds under authorized policy; funds OMF deployment                                                                      | —                                                                                             |

Two capture vectors are explicitly foreclosed. Operator capture: the ORF operator does not obtain authority over technical governance merely because it produces revenue — community governance retains absolute replacement power, exactly as the dOSPO's replaceability safeguard requires. Customer capture: a commercial member or enterprise customer does not acquire maintainer control because it pays the ecosystem. Commercial influence and technical merit must remain permanently distinguishable — the ORF extension of the OMF's Maintainer Autonomy safeguard.

#### LEGAL EXECUTION PRECEDENT

*The Polkadot Community Foundation — a Cayman Islands foundation company structured as an “unopinionated” off-chain executor of OpenGov referenda — signs contracts, executes fiat payroll, and holds assets while DOT holders retain 100% governance authority, including the power to withhold funding, appoint directors, or dissolve the entity. This is the strongest available template for the ORF Neutral Legal Entity.*

— Polkadot referenda #1122, #1416, #1591; OS Frontiers Evidence Register, 2026



---

## Legal and Tax Compliance Rules (U.S. Tax-Exempt Entities)

Two compliance constraints deserve explicit statement because they are routinely ignored in ecosystem revenue proposals. Both are specific to United States federal tax law; ecosystems incorporating elsewhere require separate analysis under their own jurisdictions, and a Cayman or European wrapper will face a materially different question set.

First, Unrelated Business Income Tax (UBIT). For U.S. tax-exempt entities, commercial service revenue is generally subject to UBIT unless the activity is substantially related to the exempt purpose. Critically, applying the proceeds to an exempt purpose does not by itself make the underlying revenue substantially related — funding maintainers with subscription income does not exempt the subscription income. Neutral legal entities must therefore maintain proper tax reserves against commercial receipts.

Second, Recoverable Mission Funding (RMF). Arrangements that finance open source projects with potential repayment or revenue participation should not be characterized as Program-Related Investments without transaction-specific tax analysis confirming the requirements of IRC §4944(c) and applicable Treasury Regulations — in particular that the investment's primary purpose is the advancement of an exempt purpose and that the production of income or appreciation of property is not a significant purpose.

Operating expenses — sales, counsel, filing fees, tax compliance, support delivery — are paid by the Neutral Legal Entity from gross receipts before net proceeds flow to the treasury, which is why net-contribution accounting is not optional bookkeeping but structural necessity.



## The Service Capacity Test

An ecosystem MAY NOT offer or sell Extended Lifecycle Support or maintenance SLAs until passing a formal Service Capacity Test demonstrating all five conditions:

|                                                |                                                                                                                   |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>1. Contracted Maintainer Capacity</b>       | Dedicated engineering hours committed under contract — in practice, an operating OMF Maintainer Retainer program. |
| <b>2. Triage &amp; Support Infrastructure</b>  | 24/7 or defined intake, ticketing, and severity classification.                                                   |
| <b>3. Escalation &amp; Backport Procedures</b> | Documented patch release pipelines and cross-team escalation paths.                                               |
| <b>4. Liability &amp; Insurance Coverage</b>   | Contractual liability caps, insurance, and legal defense arrangements via the Neutral Legal Entity.               |
| <b>5. Wind-Down Operating Reserve</b>          | A reserved customer refund pool (six months of obligations) held against outstanding commitments.                 |

An ecosystem failing the test is restricted to Open Infrastructure Assurance — risk intelligence, reporting, and coordination without contractual remediation guarantees. The test is the framework's structural answer to the most dangerous failure mode in commercial open source: liability without capacity.



## Measuring Replenishment: The Six Ratios

Metrics without decision rules are reports, not governance tools. ORF quantifies ecosystem replenishment health through six standardized ratios. Five feed a specific hard gate; the sixth is a mandatory disclosure that triggers portfolio review rather than failure.

| Ratio                                         | Definition                                         | Measures                                                                                                                                                      |
|-----------------------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Incremental Earned Coverage (IECR)</b>  | Net earned revenue ÷ cost floor                    | How much of the baseline floor is covered by newly earned commercial revenue — subscriptions, dues, certifications.                                           |
| <b>2. Structural Protocol Coverage (SPCR)</b> | Net protocol revenue ÷ cost floor                  | Coverage generated through non-inflationary structural network activity: fee splits and sequencer contributions.                                              |
| <b>3. Portfolio Coverage (PCR)</b>            | (earned + protocol + yield + pledges) ÷ cost floor | Total recurring non-inflationary net replenishment against the verified cost floor. PCR = 1.0 is the Gate 3 threshold.                                        |
| <b>4. Stress Coverage (SCR)</b>               | Stressed inflows ÷ austerity floor                 | Whether critical infrastructure remains financeable after simultaneous shocks — the Gate 6 threshold.                                                         |
| <b>5. Revenue Concentration (RCR)</b>         | Largest single payer ÷ total net inflows           | Single-payer dependency risk. RCR must remain at or below 0.25 — the Gate 5 threshold.                                                                        |
| <b>6. Mechanism Concentration (MCR)</b>       | Largest single mechanism ÷ total net inflows       | Source dependency risk. Disclosed every quarter; an MCR above 0.50 triggers mandatory portfolio review and an explicit stress justification. Not a hard gate. |

### INTERPRETIVE NOTE ON RCR

*RCR measures counterparty concentration, not the share represented by a revenue family. A protocol fee mechanism may account for a large portion of inflows without constituting a single commercial payer; dependency on that mechanism is analyzed instead through correlation-class diversity (Gate 4) and stress testing (Gate 6). This formulation removes the ambiguity between payer concentration and source concentration while preserving the Gate 5 safeguard. Where a structural mechanism does have a dominant identifiable counterparty — as the Base/Optimism case demonstrated — that counterparty is a payer, and RCR applies in full.*

### Why RCR Alone Is Insufficient

RCR and MCR answer two different questions, and a portfolio can pass the first while failing the second. An ecosystem deriving 70% of its economics from a single protocol fee mechanism has no dominant commercial payer — RCR may sit comfortably below 0.25 — yet it is overwhelmingly dependent on one source of value. Correlation classes (Gate 4) and stress testing (Gate 6) capture part of that exposure indirectly; MCR states it



directly. In the Tier 1 scenario modeled in Section 15, protocol fees represent 47.4% of total net inflows: an acceptable but review-triggering concentration that the payer-based RCR would never surface.

The Evidence Register additionally defines a composite Exit Risk Score — the product of payer concentration, contract durability, exit rights, economic correlation, and enforceability — derived directly from the Optimism/Base stack transition, for evaluating whether nominally structural inflows conceal counterparty exit exposure.



## The Eight Hard Gates for Self-Sustainability

ORF intentionally rejects additive maturity scoring. An ecosystem must not be able to compensate for insolvency with excellent governance documentation, or for customer concentration with revenue growth. To claim self-sustaining status, all eight gates must hold simultaneously.

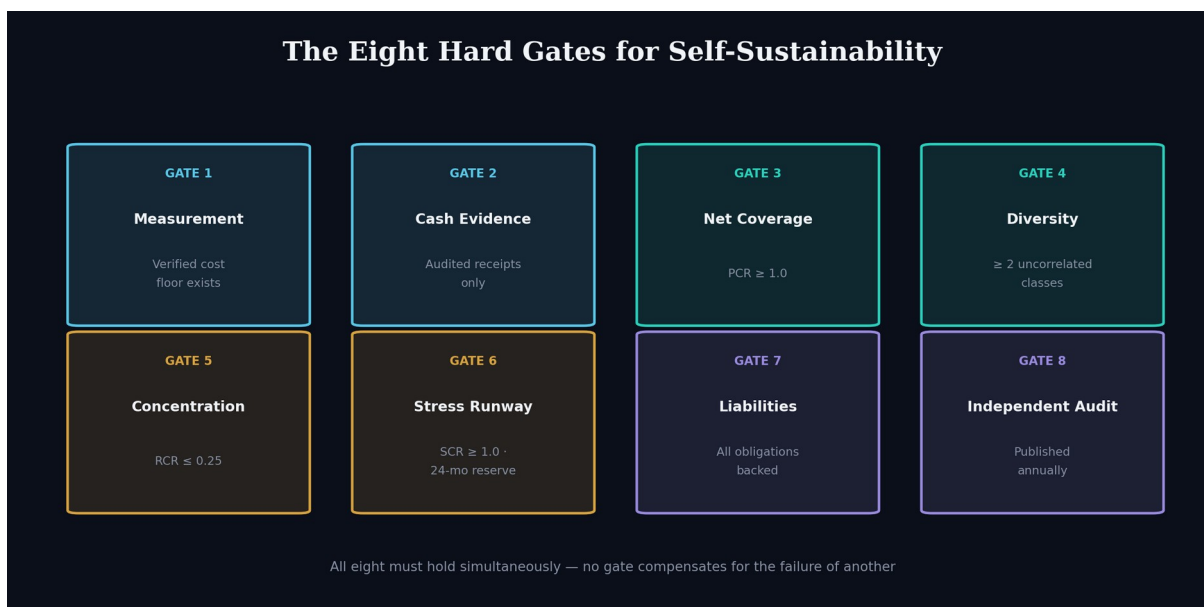


Figure 6: The Eight Hard Gates — no gate compensates for the failure of another.

|                                       |                                                                                                                                                                                             |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Gate 1 — Measurement</b>           | An empirically verified annual maintenance cost floor exists for the defined infrastructure scope.                                                                                          |
| <b>Gate 2 — Cash Evidence</b>         | Inflow metrics reflect audited cash or stablecoin receipts actually received. Forecasts, unsigned commitments, un-realized token pledges, and hypothetical demand do not qualify.           |
| <b>Gate 3 — Net Coverage</b>          | Recurring non-inflationary net inflows equal or exceed 100% of the baseline floor ( $PCR \geq 1.0$ ).                                                                                       |
| <b>Gate 4 — Multi-Class Diversity</b> | Inflows originate from at least two materially uncorrelated Revenue Correlation Classes, with independence established through the three-step procedure in Section 13 rather than asserted. |
| <b>Gate 5 — Concentration Limit</b>   | No single commercial payer accounts for more than 25% of recurring net inflows ( $RCR \leq 0.25$ ).                                                                                         |
| <b>Gate 6 — Stress Runway</b>         | Critical operations remain financeable under the defined compound stress scenario, with at least 24 months of liquid operating reserve preserved ( $SCR \geq 1.0$ ).                        |



|                                    |                                                                                                                                                         |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Gate 7 — Liability Coverage</b> | All customer SLAs, maintainer retainers, refund obligations, and operational liabilities are fully backed by allocated capacity and wind-down reserves. |
| <b>Gate 8 — Independent Audit</b>  | Annual financial and operational audits are published publicly by an independent third party.                                                           |

These gates change what “sustainable” means. A treasury holding a large token reserve is not automatically sustainable. An ecosystem with one highly profitable customer is not automatically sustainable. A protocol with high current fee revenue is not automatically sustainable. ORF requires coverage, evidence, diversification, resilience, liability control, and auditability at the same time — because each gate blocks a documented failure mode that the others cannot.



## Revenue Correlation Classes

Diversification in ORF is an analysis of economic dependence, not a count of instruments. The framework currently identifies six correlation classes.

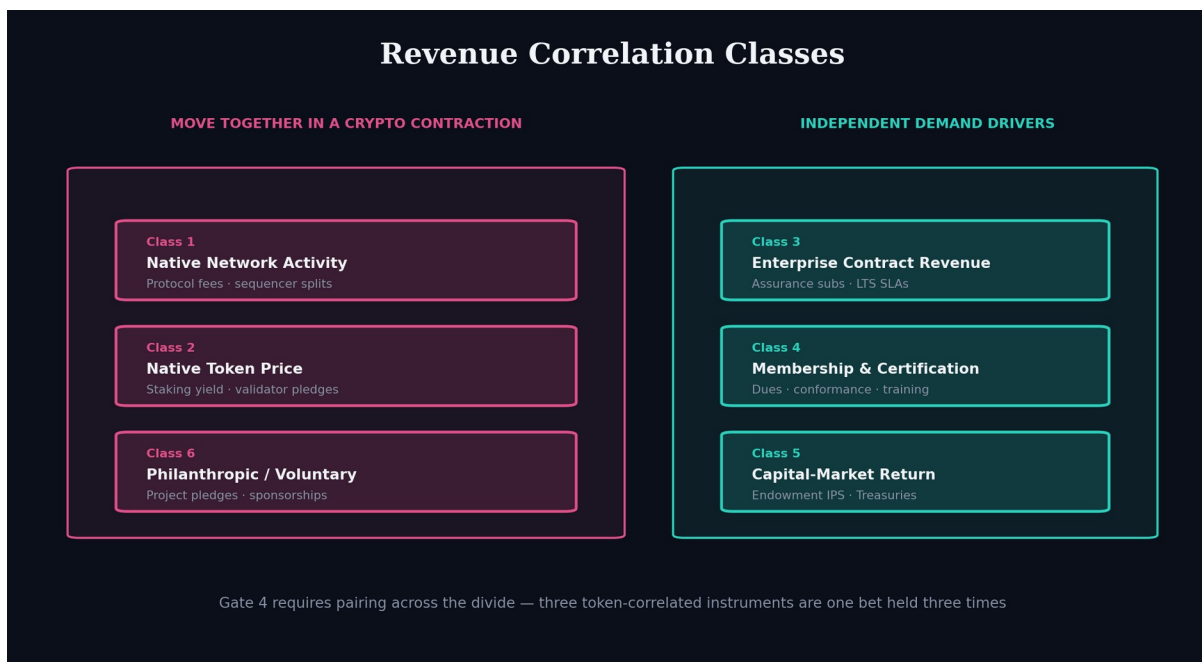


Figure 7: Revenue Correlation Classes — Gate 4 requires pairing across the divide.

| Class   | Name                        | Example Inflows                                                        |
|---------|-----------------------------|------------------------------------------------------------------------|
| Class 1 | Native Network Activity     | Protocol transaction fees, sequencer profit splits                     |
| Class 2 | Native Token Price          | Staking yield, validator pool pledges, token-denominated contributions |
| Class 3 | Enterprise Contract Revenue | Assurance subscriptions, LTS SLAs                                      |
| Class 4 | Membership & Certification  | Consortium dues, provider certification, training                      |
| Class 5 | Capital-Market Return       | Endowment IPS yield, US Treasury sleeves                               |
| Class 6 | Philanthropic / Voluntary   | Ecosystem dApp pledges, public grants                                  |

### Determining Independence

Gate 4 turns a judgment about correlation into a hard sustainability condition, so the judgment cannot be left informal. ORF requires independence to be established through a three-step procedure, applied in order and documented in the quarterly audit.



|                                             |                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Step 1 — Historical correlation</b>      | Where at least eight quarters of receipts exist for both classes, compute the historical correlation of quarterly net inflows. Classes correlating above 0.6 are treated as a single class for Gate 4 purposes regardless of their nominal classification.                                                                                                                      |
| <b>Step 2 — Common risk-driver analysis</b> | Where insufficient history exists — the normal case for a new portfolio — identify the primary demand driver of each class and document whether the two share one. Native token price, on-chain transaction volume, and ecosystem sentiment are the recurring shared drivers in Web3; enterprise procurement cycles and capital-market rates are typically independent of them. |
| <b>Step 3 — Stress co-movement test</b>     | In all cases, the compound stress scenario must apply shocks to both classes simultaneously and demonstrate that the portfolio survives. A pairing that fails co-movement testing does not satisfy Gate 4 even if Steps 1 and 2 suggest independence.                                                                                                                           |

The decisive insight: Classes 1, 2, and partially 6 tend to move together in a crypto contraction — network activity falls, token price falls, and goodwill pledges shrink simultaneously. A portfolio holding protocol fees, staking yield, and validator pledges holds one bet three times. Gate 4 is satisfied only by combining materially independent classes — the canonical pairing being structural network revenue (Class 1) with enterprise contract revenue (Class 3), whose demand driver is corporate risk management rather than market sentiment.



## Evidence: The P0–P5 and D0–D5 Scales

ORF distinguishes evidence that a mechanism works somewhere from evidence that a specific ecosystem can deploy it. These are different questions with different evidence requirements, so ORF measures them on two separate scales rather than overloading one. External precedent establishes plausibility; deployment evidence establishes local viability.

The separation matters practically. An external observer can establish that a mechanism operates at scale from public disclosure and adoption counts, but cannot usually establish its net profitability after all operating costs. A local operator can and must establish exactly that. Collapsing both onto a single scale produced the incoherence that an earlier draft of this framework contained: CNCF conformance rated “D4 Scaled” on the strength of ninety-plus certified offerings, even though the D4 definition requires positive net contribution after all operational costs — a criterion CNCF adoption counts do not evidence.

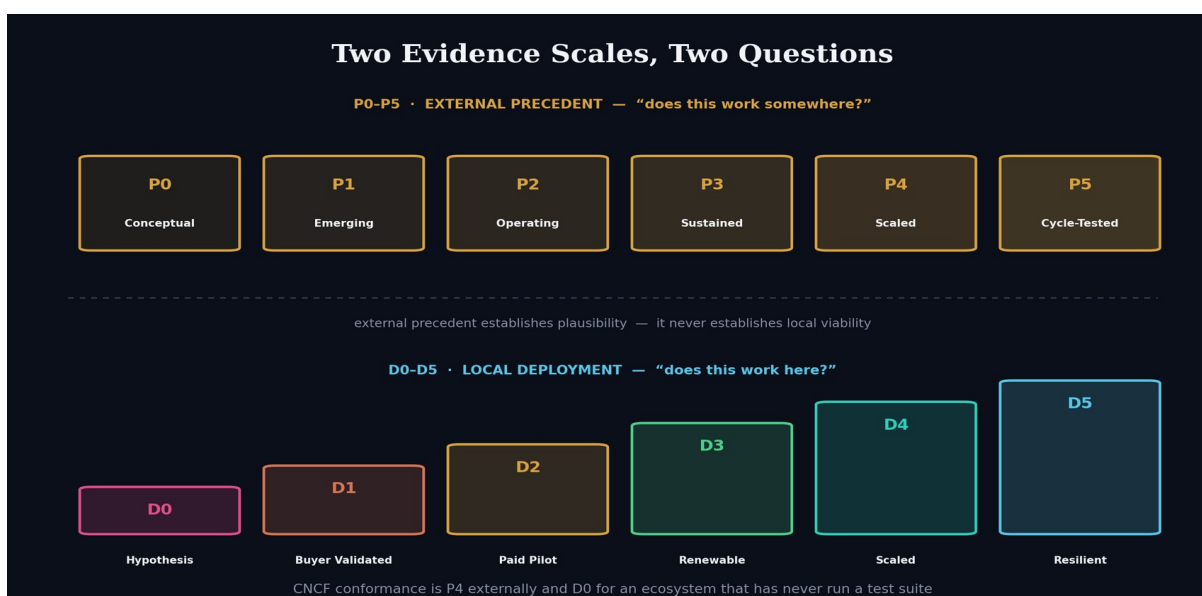


Figure 8: Two evidence scales, two questions — precedent is not proof.

### External Precedent Scale (P0–P5)

| Status | Stage      | Qualification Criteria                                                               |
|--------|------------|--------------------------------------------------------------------------------------|
| P0     | Conceptual | Proposed mechanism with no known production implementation anywhere.                 |
| P1     | Emerging   | Early implementation by at least one operator; limited public disclosure of results. |
| P2     | Operating  | Running in production with publicly disclosed receipts or adoption data.             |
| P3     | Sustained  | Multi-year operation with evidence of renewal or repeat participation.               |
| P4     | Scaled     | Large, diversified operation across multiple adopters or customers.                  |



| Status | Stage        | Qualification Criteria                                                             |
|--------|--------------|------------------------------------------------------------------------------------|
| P5     | Cycle-Tested | Continuous operation through a full market cycle with disclosed financial results. |

### Local Deployment Scale (D0-D5)

| Status | Stage           | Qualification Criteria                                                                      |
|--------|-----------------|---------------------------------------------------------------------------------------------|
| D0     | Hypothesis      | Theoretical concept; no buyer interviews or technical specifications completed.             |
| D1     | Buyer Validated | Documented buyer interviews and Letters of Intent confirming willingness-to-pay.            |
| D2     | Paid Pilot      | Production pilot executing signed contracts with real cash or stablecoin receipts.          |
| D3     | Renewable       | Paid customer renewals executed over a full multi-quarter contract cycle.                   |
| D4     | Scaled          | Diversified customer base delivering positive net contribution after all operational costs. |
| D5     | Resilient       | Multi-year operational history surviving a complete crypto or macro market downturn.        |

The paired scales prevent the most common failure in ecosystem economic design: finding an example somewhere else and treating its existence as proof the mechanism will work locally. CNCF certification is P4 as an external precedent and D0 for an ecosystem that has never written a conformance suite. The two facts coexist without contradiction, and only the second determines what may be promised to governance.



## Feasibility: The Tier 1 Scenario Model

The OS Frontiers repository includes a quantitative feasibility scenario testing the full architecture against an illustrative mid-sized production ecosystem with a \$3.0M annual maintenance cost floor: core consensus and client maintenance (\$1.6M, 53.3%), SDK and tooling maintenance (\$600K, 20.0%), security audits and vulnerability triage (\$400K, 13.3%), and program operations, legal, and administration (\$400K, 13.3%).

### The Defensible Initial Commercial Pilot

Restricting the launch portfolio to high-probability, buyer-validated enterprise instruments:

| Instrument                        | Assumption              | Gross       | Overhead | Net Contribution |
|-----------------------------------|-------------------------|-------------|----------|------------------|
| Open Infrastructure Assurance     | 5 customers @ \$75K/yr  | \$375,000   | 20%      | \$300,000        |
| Sustaining Consortium Memberships | 10 members @ \$50K/yr   | \$500,000   | 15%      | \$425,000        |
| Certified Ecosystem Provider      | 10 offerings @ \$25K/yr | \$250,000   | 22%      | \$195,000        |
| Defensible Commercial Net Total   | —                       | \$1,125,000 | —        | \$920,000 / yr   |

This yields an Incremental Earned Coverage Ratio of 30.7%. The result matters precisely because it does not claim self-sustainability: a credible initial commercial launch materially reduces treasury dependence — by nearly a third — without eliminating it. Overpromising is itself a failure mode; the model refuses to.

### The Full Multi-Family Scenario

Adding governance-legitimized structural protocol fees (\$1.5M gross, 2% overhead, \$1.47M net) and governed capital yield (\$750K gross, 5% overhead, \$712.5K net) produces total modeled net replenishment of \$3,102,500 against the \$3.0M floor — a Portfolio Coverage Ratio of 103.4%, satisfying Gate 3, with no commercial customer exceeding 15% of net inflows and four correlation classes represented.

Two model assumptions must be stated explicitly rather than left implicit, because both are load-bearing and both are the kind of omission this framework criticises elsewhere.

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Capital principal requirement | The \$750,000 gross capital-yield sleeve is not free-standing. At the 3–5% real yield assumed in Family E, it requires approximately \$15M–\$25M of productive principal already under management, before operating costs, reserves, tax, and volatility. An ecosystem without that principal cannot include this line, and a scenario that omits the requirement commits precisely the Endowment Fantasy failure mode catalogued in Section 18. |
| Opening liquid reserve        | Gate 6 runway is computed from a stated opening liquid operating reserve of \$5,130,000 held in stable assets, against an austerity floor of \$1,800,000 per year, or \$150,000 per month. The resulting 34.2 months is deliberately conservative: it assumes all stressed inflows cease entirely rather than continuing at their stressed levels.                                                                                               |

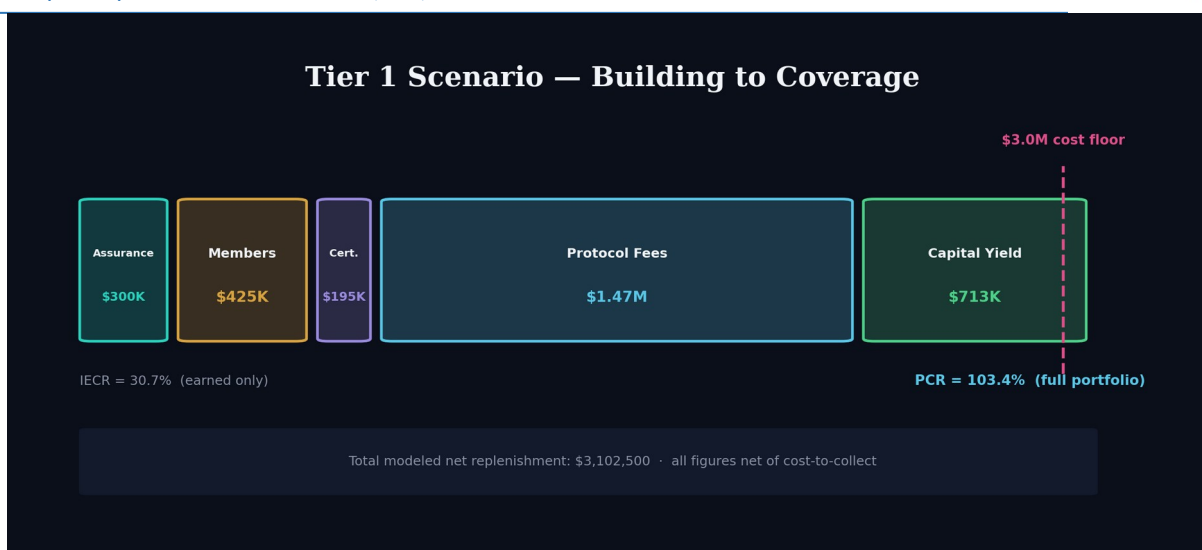


Figure 9: Tier 1 Scenario — building from defensible pilot revenue to full portfolio coverage.

## The Compound Stress Test

Four simultaneous shocks are applied: protocol fees down 50% (crypto winter), enterprise assurance down 40% (churn), membership dues down 30% (contraction), and capital yield down 40% (drawdown). Certification revenue is held flat, on the assumption that conformance contracts for the period are already executed; holding it flat is an assumption, not an omission, and an ecosystem with annually renewing certification should shock this line as well.

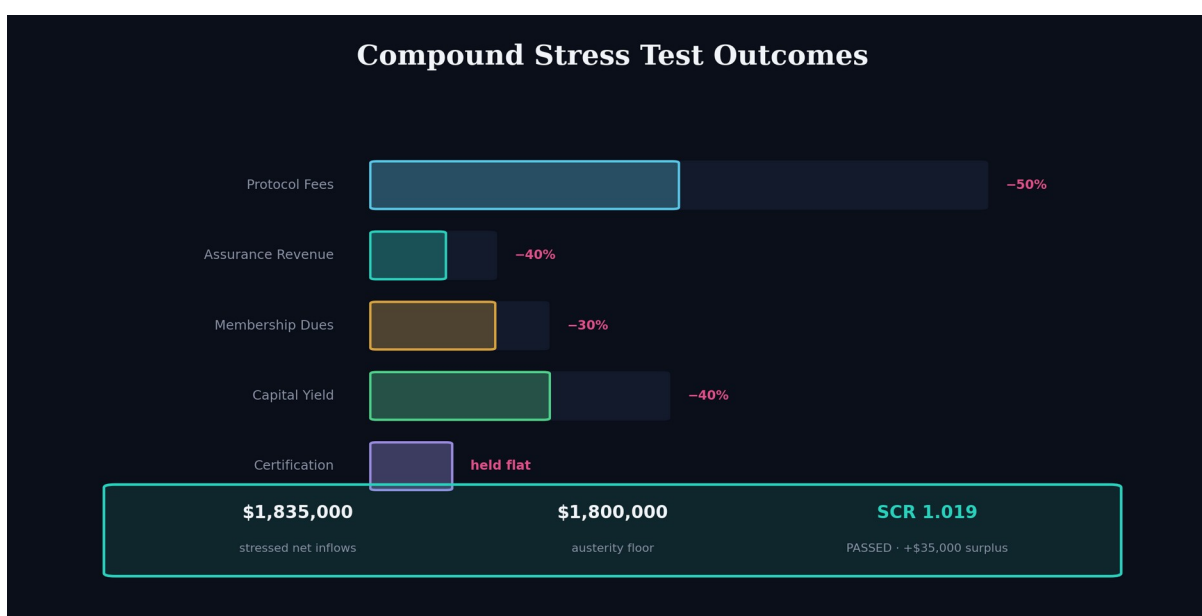


Figure 10: Compound Stress Test — stressed inflows against the austerity maintenance floor.

| Line                 | Base Net    | Shock | Stressed Net |
|----------------------|-------------|-------|--------------|
| Protocol fees        | \$1,470,000 | -50%  | \$735,000    |
| Enterprise assurance | \$300,000   | -40%  | \$180,000    |



| Line                | Base Net    | Shock     | Stressed Net |
|---------------------|-------------|-----------|--------------|
| Membership dues     | \$425,000   | -30%      | \$297,500    |
| Capital yield       | \$712,500   | -40%      | \$427,500    |
| Certification       | \$195,000   | held flat | \$195,000    |
| Stressed total      | \$3,102,500 | —         | \$1,835,000  |
| Austerity floor     | —           | —         | \$1,800,000  |
| Surplus / (deficit) | —           | —         | +\$35,000    |

The Stress Coverage Ratio is therefore 1.019 — above the Gate 6 threshold of 1.0, but only barely, and the thinness of that margin is the useful result. Combined with the \$5,130,000 opening reserve, the scenario preserves 34.2 months of runway against the austerity floor. Mechanism Concentration is 47.4%: below the 0.50 review trigger, but close enough to it that a prudent operator would document the protocol-fee dependency in the quarterly audit rather than wait for the threshold to be crossed. Under the pilot-only scenario, the ecosystem honestly reports Gate 3 failure and a named 69.3% coverage gap — which is the framework working as intended.

#### METHODOLOGICAL NOTE

*These figures are scenario parameters, not forecasts. They demonstrate how ORF should be tested mathematically — not that another ecosystem will achieve the same pricing, demand, costs, or revenue mix. Every parameter must be re-derived locally through Phase I discovery.*



## Implementation Roadmap

ORF deploys incrementally. Attempting all instruments simultaneously overloads governance capacity and multiplies failure modes — the same lesson the OMF drew from the Sovereign Tech Agency's start-modest trajectory.

| Phase                                         | Window       | Actions                                                                                                                                                                                                                                                                                   | Objective                                            |
|-----------------------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| <b>Phase I<br/>Baseline &amp; Discovery</b>   | Months 0–3   | Determine the actual maintenance cost floor. Map treasury and grant dependence. Inventory and classify current economic activity. Measure native-token exposure. Form the Neutral Legal Entity. Complete 15–25 enterprise buyer interviews. Separate revenue from routing already in use. | Not revenue — an evidence-backed economic baseline.  |
| <b>Phase II<br/>Commercial Validation</b>     | Months 3–6   | Pre-sell low-liability instruments: assurance subscriptions, sustaining membership tiers, conformance certification. Onboard 3–5 pilot customers before building expensive support operations.                                                                                            | Move candidate instruments from D0 toward D1 and D2. |
| <b>Phase III<br/>Operationalization</b>       | Months 6–12  | Build repeatable commercial operations. Contract maintainers through OMF retainers where LTS is planned; pass the Service Capacity Test before any SLA. Implement net-contribution accounting, liability reserves, and the 12-field standard. Integrate transparent routing rails.        | Repeatability.                                       |
| <b>Phase IV<br/>Portfolio Diversification</b> | Months 12–24 | Evaluate additional correlation classes: governance-legitimized structural revenue, certification expansion, governed Endowment IPS, contribution pledges. Advanced risk products remain research-horizon.                                                                                | Survive the failure of any single revenue source.    |

Indicative first-year operating cost for a minimum viable ORF — a commercial lead, part-time legal and accounting via the Neutral Legal Entity, and reporting infrastructure — falls in the same \$450K–\$1.2M coordination band the dOSPO paper established, and is evaluated against the same test: net contribution after the coordination cost, not gross activity.



## The Reporting Standard

Every replenishment mechanism must generate sufficient data to answer twelve mandatory questions, published as a public Quarterly Replenishment Audit by the ORF Operator:

| #  | Field                            | Requirement                                                            |
|----|----------------------------------|------------------------------------------------------------------------|
| 1  | Gross Cash Received              | Actual USD or stablecoin receipts (un-realized token pledges excluded) |
| 2  | Collection & Service Cost        | Itemized sales, legal, tax, and support delivery overhead              |
| 3  | Net Contribution                 | Net capital transferred to the Governed Treasury                       |
| 4  | Source Classification            | Protocol, Enterprise, Voluntary, Investment, Issuance, or Routing Rail |
| 5  | Recurring vs One-Time            | Multi-year subscriptions and fees vs episodic receipts                 |
| 6  | Contract Term & Termination      | Duration and exit rights                                               |
| 7  | Payer Concentration (RCR)        | Share from the largest single payer (at or below 0.25)                 |
| 8  | Native-Token Exposure            | Correlation factor with native token market price                      |
| 9  | Customer Renewal Rate            | Verified percentage of commercial subscribers renewing                 |
| 10 | Outstanding Service Liabilities  | Unfulfilled SLA obligations and refund reserves                        |
| 11 | Restricted / Unrestricted Status | Treasury accounting classification                                     |
| 12 | Audit & Provenance Status        | Third-party verification details and transaction hashes                |

The standard exists to make three specific deceptions impossible: relabeling treasury transfers as revenue (fields 1 and 4), disguising dilution as income (fields 4 and 8), and reporting gross activity as sustainability (fields 2, 3, and 10).



## Failure Modes

An ORF implementation can fail while generating revenue. Nine documented failure patterns map directly to the framework's safeguards.

| Failure Mode                      | Description                                                                                                                                 | Blocked By                                             |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| <b>Treasury Relabeling</b>        | Calling existing grant or treasury transfers “revenue”                                                                                      | Gate 2; reporting fields 1, 4                          |
| <b>Token Dilution as Income</b>   | Counting issuance as captured external value                                                                                                | Instrument A.4 policy; Gate 3                          |
| <b>Gross-Revenue Accounting</b>   | Ignoring sales, delivery, tax, insurance, and legal costs                                                                                   | Net-contribution principle; fields 2–3                 |
| <b>Customer Capture</b>           | Commercial payers converting support into technical influence                                                                               | Institutional separation (Section 9)                   |
| <b>Revenue Concentration</b>      | One major payer, or one dominant mechanism, creating apparent strength with severe exit risk — the Optimism/Base lesson                     | Gate 5 (RCR); MCR disclosure; Exit Risk Score          |
| <b>Liability Without Capacity</b> | Selling response guarantees before establishing maintainer capacity                                                                         | Service Capacity Test; Gate 7                          |
| <b>False Diversification</b>      | Multiple token-correlated streams collapsing together                                                                                       | Correlation classes; Gate 4                            |
| <b>Endowment Fantasy</b>          | Treating capital yield as a bootstrap for under-capitalized treasuries, or modelling a yield line without stating its principal requirement | Family E reality check; mandatory principal disclosure |
| <b>Routing Confusion</b>          | Counting payment rails and allocation engines as revenue                                                                                    | Strict Functional Separation; field 4                  |

These failure modes explain why ORF pairs its instrument catalog with hard gates and governance safeguards rather than merely enumerating monetization ideas. Every safeguard in this paper is a response to a failure that has already occurred somewhere.



## Research Status and the Road Ahead

ORF is a Stage 0 Research Candidate within the Open Source Frontiers validation lifecycle: Stage 0 (Research Candidate), Stage 1 (Independent Peer Review), Stage 2 (Piloted Precursors), Stage 3 (Validated Production). That designation is deliberate, and it mirrors the epistemic honesty this series has attempted throughout: the dOSPO paper conceded it should be dismantled when unjustified; the OMF paper published its claims with confidence ratings; the ORF declines to declare the sustainability problem solved.

The architecture synthesizes mechanisms with verified production precedents. The integrated system has not yet demonstrated multi-quarter production sustainability as a complete framework. For ORF specifically, validation requires more than architectural review. It requires real buyers, real contracts, real collection costs, real renewals, real maintainers, real liabilities, real market downturns — and longitudinal evidence that replenishment reduces net dependence on finite treasury capital over time.

The field is moving in ORF's direction. Optimism's revenue-linked buyback vote, ENS's endowment maturation, Tidelift's enterprise expansion under Sonar, Protocol Guild's continued donor growth, and Cardano's POSM deployment programs all indicate that individual loop segments are hardening in production. What remains missing is the connected, measured whole — and an ecosystem willing to run it honestly against the eight gates.

Until those conditions are observed, ORF should be understood as a testable architecture, not a declaration of victory.



## Adversarial Review: Critiques and Rebuttals

This section addresses the strongest objections to the ORF model. These critiques are not strawmen; they reflect arguments raised by open source purists, DAO participants, foundation operators, enterprise open source veterans, and token holders.

### **“This is monetizing open source with extra steps.”**

*Argument: However framed, ORF introduces commercial extraction into commons that should remain freely given.*

Rebuttal: ORF explicitly prohibits charging for code access and anchors every commercial instrument to independent counter-value — assurance, testing, certification, response capacity — that the payer separately wants and that costs real money to produce. The alternative to earned revenue is not a pure commons; it is permanent dependence on token issuance, treasury drawdown, and donor sentiment, all of which have documented failure records. The code remains open. The question is only whether the people maintaining it are financed by depletion or by the value they enable.

### **“Protocol fee splits are just taxes.”**

*Argument: Structural collection extracts value from network participants who never consented to fund maintenance.*

Rebuttal: Partially correct, which is why ORF refuses to treat structural collection as a sale. Fee splits change how network economics are distributed and therefore require explicit governance legitimacy — token-holder authorization through the ecosystem's own constitutional process, exactly as Cardano's treasury cut and Optimism's Superchain charter were adopted. ORF's contribution is the bright line: instruments are either sold (counter-value) or governed (structural mandate). What it prohibits is the middle path — quiet extraction framed as neither.

### **“This is just Tidelift or the Linux Foundation with a token.”**

*Argument: Everything here already exists in Web2 commercial open source.*

Rebuttal: Individually, yes — and the paper cites those precedents deliberately. What does not exist anywhere is the portfolio: correlation-aware diversification requirements, net-contribution accounting, payer concentration limits, stress-tested coverage ratios, a capacity test gating liability products, and hard non-compensable gates, operated under decentralized governance with a replaceable operator. The Evidence Register's central finding stands: among the thirteen precedents reviewed, none implements the complete closed loop. Synthesis under measurement discipline is the contribution.

### **“An endowment solves this more simply.”**

*Argument: Accumulate capital once, live on yield forever, skip the commercial machinery.*

Rebuttal: The arithmetic refuses. \$3M of annual spendable yield requires \$60M–\$100M+ of productive principal; ENS — the best-executed endowment in the space — covers roughly a fifth of its expenses from returns. For the small number of ecosystems that can capitalize at that scale, Family E is in the framework — and Section 15 now states the principal requirement behind its own modeled yield line (\$15M–\$25M for \$750K of gross yield) rather than quietly assuming it. For everyone else, the endowment answer is a decade-long fundraising project wearing the costume of a solution.

### **“Enterprises will never pay DAOs.”**

*Argument: Commercial buyers require counterparties, contracts, invoices, and liability — things decentralized ecosystems cannot provide.*



Rebuttal: Correct as stated, which is precisely why ORF makes the Neutral Legal Entity non-optional for commercial instruments. The Polkadot Community Foundation demonstrates that a neutral legal executor can sign contracts, run payroll, and carry liability while governance retains complete authority over it. Enterprises do not pay DAOs; they pay the entity the DAO governs. Tidelift's customer list — Cisco, Fannie Mae, the U.S. Air Force — demonstrates the demand exists when the counterparty problem is solved.

**“The eight gates are unreachable strict.”**

*Argument: Requiring all eight conditions simultaneously sets a bar nothing clears, making the framework decorative.*

Rebuttal: The gates define “self-sustaining,” not “worth doing.” An ecosystem at 30% coverage with honest reporting has reduced its treasury burn by nearly a third and knows exactly what remains — that is the framework succeeding. The strictness is the point: every relaxed gate corresponds to a documented collapse. A standard that a treasury-rich, single-customer, token-correlated ecosystem could pass would be a standard not worth publishing.

**“Revenue programs will capture governance.”**

*Argument: Whoever controls the money printer eventually controls the ecosystem.*

Rebuttal: This is the series' central preoccupation, answered structurally rather than culturally: the ORF operator holds no technical authority and is replaceable by governance; the Neutral Legal Entity executes but does not decide; customers buy services, never influence; the dOSPO audits both operators while holding zero custody; and independent annual audit is itself a hard gate. The goal — as the dOSPO paper put it — is not to prevent abuse forever but to make abuse legible and reversible.



## Conclusion

This series began with a governance question and ends with an economic one.

The dOSPO paper argued that decentralized ecosystems fail when responsibility dissolves under distributed authority, and proposed a bounded, replaceable coordination layer: who decides. The OMF paper argued that the sustainability crisis is an operational-architecture problem, and defined the deployment engine that turns governance mandates into sustained maintenance: how resources go out. Both frameworks presuppose capital. This paper has addressed the presupposition: how resources come back.

The answer is not a token, a smart contract, a subscription product, an endowment, a tax, or any single funding mechanism. It is a portfolio architecture: a structured method for identifying legitimate sources of recurring value, measuring their true net contribution, diversifying them across economic risk classes, protecting technical governance from financial capture, maintaining adequate service capacity, routing proceeds transparently, and determining — against hard, non-compensable gates — whether the resulting system actually finances the cost of open source maintenance.

The long-term objective is a transition from treasury to funding to maintenance to depletion, toward maintenance that sustains open infrastructure, which enables economic value, which returns as replenishment, which finances the next cycle of maintenance. A closed loop does not mean ecosystems will never again need grants, philanthropy, treasury reserves, or public-goods funding. It means those sources no longer carry the entire burden alone.

The evidence from production systems supports each segment of the loop. Optimism demonstrates that structural network revenue can finance ecosystems at scale. ENS demonstrates that canonical service revenue and governed capital can operate together. Tidelift demonstrates that enterprises pay for assurance around open infrastructure. Protocol Guild demonstrates that voluntary replenishment can be structured as a capture-resistant commons. Polkadot demonstrates that legal execution can serve decentralized authority. What no ecosystem yet demonstrates is the connected whole — and that synthesis, held to measurement discipline, is what this series now places on the table.

**Open source infrastructure does not sustain itself.**

**It survives because institutions emerge to support it — and those institutions survive only if value flows back to them.**

The dOSPO provides the mandate. The OMF provides the machinery. **The Open Replenishment Framework closes the loop.**



## About Open Source Cowboy

The Open Replenishment Framework was developed by Open Source Cowboy (opensourcecowboy.org) as the third installment of an ongoing body of work on decentralized open source governance, operations, and economic sustainability, alongside the dOSPO and OMF whitepapers and the Open Source Frontiers research repository at LF Decentralized Trust. ORF is designed to be adopted, adapted, and improved through use. We invite governance participants, foundation operators, treasury stewards, enterprise buyers, and maintainers across Web3 to engage with the framework and contribute to its validation.

## Evidence Summary

| Claim                                                                                                                                        | Primary Sources                                                                             | Confidence             |
|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------|
| Open source components appeared in 96% of commercial codebases analyzed (2024); OSS supply-side value ~\$4.15B, demand-side ~\$8.8T          | Synopsys OSSRA (2024); Hoffman, Nagle & Zhou, Harvard Business School (2024)                | High                   |
| Optimism mechanism: greater of 2.5% of Chain Revenue or 15% of Net Onchain Profit; 50% revenue-linked OP buyback approved 2026               | Optimism published documentation and governance record (2025–2026)                          | High                   |
| Cumulative Superchain ETH, OP-committed, and grantee-count aggregates                                                                        | Widely repeated in ecosystem commentary; not traced to a dated primary Optimism publication | Low — pending citation |
| Base moved to a Base-managed unified stack in 2026 while retaining a transitional Optimism relationship; exposes exit and dependency risk    | Base and Optimism public announcements; contemporaneous reporting (2026)                    | High                   |
| ENS: \$18.22M 2025 revenue; ~\$93.4M endowment AUM and >\$8M net DeFi returns since inception under May 2026 EP6.46; ~20.3% expense coverage | ENS DAO governance forum reporting; EP6.46 (May 2026); endowment manager review             | High                   |
| Tidelift enterprise assurance with named customers; SBOM-based maintainer payments; acquired by Sonar                                        | Tidelift published materials (2024); Sonar acquisition press release                        | High                   |
| Protocol Guild: \$7.2M from 6,202 donors (2025 annual report); 196 members (Q2 2026 audit); \$80M+ commitments                               | Protocol Guild 2025 annual report; Q2 2026 quarterly audit                                  | High                   |
| Polkadot PCF as neutral Cayman legal executor; ~\$70.6M 2025 treasury spend                                                                  | Polkadot OpenGov referenda #1122, #1416, #1591; treasury reporting                          | High                   |
| Octant: 100,000 ETH staked; ~460 ETH (~\$1.7M) Epoch 8 distribution                                                                          | Golem Foundation / Octant documentation (2025)                                              | High                   |
| Cardano POSM: 5.885M ADA 2025 budget; \$300K bounty pool fully utilized by Jul 2026                                                          | Intersect POSM reporting; on-chain budget action (2025–2026)                                | High                   |
| CNCF Certified Kubernetes: 90+ certified offerings (external scale, P4); payment buys                                                        | CNCF conformance program documentation                                                      | High                   |



| Claim                                                                                                                                      | Primary Sources                                                             | Confidence |
|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|------------|
| testing, not results                                                                                                                       |                                                                             |            |
| Drips, Superfluid, and Deep Funding are routing and allocation infrastructure, not revenue                                                 | Protocol documentation; OS Frontiers Evidence Register (2026)               | High       |
| Among the 13 precedents reviewed, none implements the complete dOSPO → OMF → ORF closed loop                                               | OS Frontiers Evidence Register comparative analysis of 13 precedents (2026) | Medium     |
| Tier 1 scenario: \$920,000 pilot net (IECR 30.7%); PCR 103.4%; SCR 1.019; MCR 47.4%; 34.2-month runway on a stated \$5.13M opening reserve | OS Frontiers Tier 1 Feasibility Model (illustrative scenario, not forecast) | Medium     |
| ORF as a portable portfolio architecture across ecosystems                                                                                 | Synthesized from comparative analysis; no single source                     | Medium     |



## Bibliography

---

- Cardano / Intersect (2026). The Paid Open Source Model (POSM). <https://www.intersectmbo.org/news/the-paid-open-source-model>
- Cardano Docs. Monetary policy: reward pot composition and treasury allocation. <https://docs.cardano.org/about-cardano/explore-more/monetary-policy>
- CNCF. Certified Kubernetes Conformance Program. <https://www.cncf.io/training/certification/software-conformance/>
- CoinDesk (2026). Coinbase's Base moves away from Optimism's OP Stack in major tech shift. <https://www.coindesk.com/business/2026/02/18/coinbase-s-base-moves-away-from-optimism-s-op-stack-in-major-tech-shift>
- ENS DAO (2026). EP6.46: 2026 Endowment Investment Policy Update. <https://discuss.ens.domains/t/6-46-social-2026-endowment-investment-policy-update/22106>
- ENS DAO (2026). KPK 2025 review for the ENS Endowment. <https://discuss.ens.domains/t/kpk-2025-review-for-the-ens-endowment/21829>
- Golem Foundation (2025). Octant documentation and epoch reports. <https://octant.build/>
- Internal Revenue Service. Publication 598: Tax on Unrelated Business Income of Exempt Organizations. <https://www.irs.gov/publications/p598>
- Internal Revenue Service. Program-Related Investments guidance under IRC §4944(c). <https://www.irs.gov/pub/irs-wd/202123004.pdf>
- Hoffman, M., Nagle, F., & Zhou, Y. (2024). The Value of Open Source Software. Harvard Business School Working Paper 24-038. <https://www.hbs.edu/faculty/Pages/item.aspx?num=65230>
- LF Decentralized Trust — Open Source Frontiers Lab (2026). os-frontiers repository: ORF specification, Instrument Catalog, Governance Rules, Tier 1 Feasibility Model, Evidence Register, Fork Resistance Analysis. <https://github.com/LF-Decentralized-Trust-labs/os-frontiers>
- Linux Foundation. Membership and project hosting. <https://www.linuxfoundation.org/about/join>
- Optimism (2025). How (and why) the Superchain drives fees to the Optimism Collective. [https://optimism.io/blog/how-\(and-why\)-the-superchain-drives-fees-to-the-optimism-collective](https://optimism.io/blog/how-(and-why)-the-superchain-drives-fees-to-the-optimism-collective)
- Optimism (2026). OP token buybacks. <https://optimism.io/blog/op-token-buybacks>
- Polkadot OpenGov (2025–2026). Referenda #1122 (Anemoy Liquid Treasury Fund), #1416 (PCF contracts), #1591 (PCF renewal). <https://polkadot.polkassembly.io/>
- Protocol Guild (2026). Annual Report 2025. <https://www.protocolguild.org/blog/20260129-annual-report-2025>
- Protocol Guild (2026). Q2 2026 Quarterly Audit. <https://www.protocolguild.org/blog/20260604-Q2-quarterly-audit>
- Protocol Guild (2023–2026). Documentation and vesting contract. <https://protocol-guild.readthedocs.io/>
- Sonar (2024). Sonar to acquire Tidelif. <https://www.sonarsource.com/company/press-releases/sonar-to-acquire-tidelift/>
- Synopsys (2024). Open Source Security and Risk Analysis (OSSRA) report. <https://www.synopsys.com/blogs/software-security/software-bill-of-materials-bom.html>
- Taylor, C. & McCutcheon, T. (2026). The Decentralized Open Source Program Office (dOSPO): Sustaining Open Source in Web3 Ecosystems. Open Source Cowboy Consulting. March 3, 2026.
- Taylor, C. (2026). The Open Maintenance Framework (OMF): Operational Architecture for Sustaining Open Source in Web3. Open Source Cowboy Consulting. March 7, 2026.
- Tidelift (2024). State of the Open Source Maintainer Report; enterprise assurance documentation. <https://tidelift.com/>
-

